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Processing apparatus for integrating a semiconductor wafer and an annular frame through a tape

Abstract

A processing apparatus includes a wafer table that supports a wafer, a frame table that supports an annular frame, a first tape pressure bonding unit that includes a first pressure bonding roller for executing pressure bonding of a tape to the annular frame, and a second tape pressure bonding unit that includes a second pressure bonding roller for executing pressure bonding of the tape of the tape-attached annular frame to a front surface or a back surface of the wafer. A first heating unit is disposed in one of or both the frame table and the first pressure bonding roller, while a second heating unit is disposed in one of or both the wafer table and the second pressure bonding roller.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2007/0227655	12/2006	Kajiyama	438/690	H01L 21/3043
2007/0277934	12/2006	Kawashima	156/247	H01L 21/67132
2008/0011412	12/2007	Tsujimoto	156/193	H01L 24/29
2008/0023149	12/2007	Yamamoto	156/515	H01L 21/67132
2008/0038903	12/2007	Yamamoto	438/464	H01L 21/68785
2010/0055877	12/2009	Kajiyama et al.	N/A	N/A
2013/0203241	12/2012	Nakata et al.	N/A	N/A
2014/0073067	12/2013	Uchida	N/A	N/A
2020/0083104	12/2019	Harada	N/A	H01L 21/3043
2022/0189800	12/2021	Uchihio et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2010062375	12/2009	JP	N/A
2013161863	12/2012	JP	N/A
2017005033	12/2016	JP	N/A
2019201016	12/2018	JP	N/A
2019201049	12/2018	JP	N/A
2021034492	12/2020	JP	N/A

OTHER PUBLICATIONS

Uchiho, Takashi, U.S. Appl. No. 17/455,136, filed Nov. 16, 2021. cited by applicant
Office Action issued by the Patent Office of Japan in corresponding Patent Application Publication No. 2021-170673, dated Apr. 11, 2025. cited by applicant

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a processing apparatus that automatically executes work of integrating a wafer and an annular frame through a tape.

Description of the Related Art

(2) A wafer having a device region and an outer circumferential surplus region formed on a front surface thereof, the device region being a region in which multiple devices such as integrated circuits (ICs) and large-scale integrated (LSI) circuits are formed in respective areas demarcated by multiple intersecting planned dividing lines, the outer circumferential surplus region surrounding the device region, is formed into a desired thickness by being ground on a back surface thereof. Thereafter, the wafer is divided into individual device chips by a dicing apparatus or a laser processing apparatus. The respective device chips obtained by the dividing are used for pieces of electrical equipment such as mobile phones and personal computers.

(3) Before being processed by the dicing apparatus or the laser processing apparatus, the wafer is positioned in an opening part of an annular frame for housing the wafer therein, and is supported by the annular frame through an adhesive tape.

(4) However, when the adhesive tape is stuck to the front surface or the back surface of the wafer, there is a fear that part of an adhesive layer remains on the wafer and lowers the quality of the device. Thus, the present assignee has proposed a wafer processing method in which a wafer is processed by being supported by an annular frame through a thermocompression bonding tape that does not have an adhesive layer (for example, thermoplastic synthetic resin tape of a polyolefin-based resin, a polyester-based resin, or the like) (for example, refer to Japanese Patent Laid-open No. 2019-201016 and Japanese Patent Laid-open No. 2019-201049). However, there is a problem that the productivity is low because integrating the wafer and the annular frame through the thermocompression bonding tape is manually executed.

(5) Further, the present assignee has proposed the following technique. Predetermined processing is executed for a wafer in a state in which a ring-shaped reinforcing part is left in a region on a back surface of the wafer corresponding to an outer circumferential surplus region in order to make conveyance of the ground wafer easy. Thereafter, a dicing tape is stuck to the back surface of the wafer, and the wafer is supported by an annular frame. Then, the ring-shaped reinforcing part is removed from the wafer (for example, refer to Japanese Patent Laid-open No. 2010-62375).

(6) However, there is the following problem. Specifically, it is difficult to integrate the wafer with the annular frame by sticking the tape to the back surface of the wafer on which the ring-shaped reinforcing part in a protrusion shape is formed in a region corresponding to the outer circumferential surplus region. In addition, it is difficult to cut the ring-shaped reinforcing part and remove it from the wafer. Thus, the productivity is low.

SUMMARY OF THE INVENTION

(7) Thus, an object of the present invention is to provide a processing apparatus that can automatically execute work of integrating a wafer and an annular frame through a tape even when the wafer has a ring-shaped reinforcing part in a protrusion shape formed in a region on a back surface of the wafer corresponding to an outer circumferential surplus region.

(8) In accordance with an aspect of the present invention, there is provided a processing apparatus including a wafer cassette table on which a wafer cassette having multiple wafers housed therein is placed, a wafer carrying-out unit that carries a wafer out from the wafer cassette placed on the wafer cassette table, a wafer table that supports the wafer carried out by the wafer carrying-out unit, and a frame housing unit that houses multiple annular frames therein, each annular frame having an opening part for housing the wafer therein. The processing apparatus also includes a frame carrying-out unit that carries the annular frame out from the frame housing unit, a frame table that supports the annular frame carried out by the frame carrying-out unit, a first tape pressure bonding

unit that is disposed above the frame table and includes a first pressure bonding roller for executing pressure bonding of a tape to the annular frame, and a tape-attached frame conveying unit that conveys, to the wafer table, the annular frame to which the tape is pressure-bonded, and places the tape-attached annular frame on the wafer table in such a manner that a front surface or a back surface of the wafer supported by the wafer table is positioned in the opening part of the annular frame. The processing apparatus also includes a second tape pressure bonding unit that includes a second pressure bonding roller for executing pressure bonding of the tape of the tape-attached annular frame to the front surface or the back surface of the wafer, frame unit carrying-out means that carries, out from the wafer table, a frame unit in which the tape of the tape-attached annular frame and the front surface or the back surface of the wafer are pressure-bonded together by the second tape pressure bonding unit, and a frame cassette table on which a frame cassette that houses the frame unit therein is placed. A first heating unit is disposed in one of or both the frame table and the first pressure bonding roller, while a second heating unit is disposed in one of or both the wafer table and the second pressure bonding roller. Either an adhesive tape that has an adhesive layer laid on a sheet or a thermocompression bonding tape that does not have an adhesive layer on a sheet is selectively used as the tape.
(9) Preferably, in the processing apparatus, a ring-shaped reinforcing part having a protrusion shape is formed in a region on the back surface corresponding to an outer circumferential surplus region of the wafer. In addition, the processing apparatus further includes a reinforcing part removing unit that cuts and removes the ring-shaped reinforcing part from the wafer of the frame unit carried out by the frame unit carrying-out means and ring-free unit carrying-out means that carries a ring-free unit resulting from the removal of the ring-shaped reinforcing part out from the reinforcing part removing unit.
(10) Preferably, the first tape pressure bonding unit includes a roll tape support part that supports a roll tape where the tape that is yet to be used is wound, a tape take-up part that takes up the tape that has been used, a tape pull-out part that pulls out the tape from the roll tape, the first pressure bonding roller that executes pressure bonding of the tape pulled out to the annular frame, and a cutting part that cuts the tape protruding to an outer circumference of the annular frame along the annular frame.
(11) Preferably, when the tape is a thermocompression bonding tape, the first heating unit is actuated to heat one of or both the frame table and the first pressure bonding roller and execute thermocompression bonding of the thermocompression bonding tape to the annular frame.
(12) Preferably, the second tape pressure bonding unit includes an upper chamber disposed above the wafer table, a lower chamber in which the wafer table is housed, a raising-lowering mechanism that raises and lowers the upper chamber and generates a closed state in which the upper chamber is brought into contact with the lower chamber and an opened state in which the upper chamber is separated from the lower chamber, a vacuum part that sets the upper chamber and the lower chamber to a vacuum state in the closed state, and an opening-to-atmosphere part that opens the upper chamber and the lower chamber to atmosphere. Further, in a state in which the tape of the tape-attached annular frame is positioned to the front surface or the back surface of the wafer supported by the wafer table, the upper chamber and the lower chamber are set to a vacuum state while the closed state is kept through actuating the raising-lowering mechanism, and pressure bonding of the tape of the tape-attached annular frame to the front surface or the back surface of the wafer is executed by the second pressure bonding roller disposed in the upper chamber.
(13) Preferably, when the tape is a thermocompression bonding tape, the second heating unit is actuated to heat one of or both the wafer table and the second pressure bonding roller and execute thermocompression bonding of the thermocompression bonding tape to the front surface or the back surface of the wafer.
(14) Preferably, a camera is disposed in the upper chamber of the second tape pressure bonding unit, and whether an exposed surface of the wafer supported by the wafer table is the front surface

or the back surface is detected before the tape-attached annular frame is conveyed.
(15) Preferably, when the exposed surface of the wafer supported by the wafer table is the front surface, the camera acquires an identification (ID) displayed on the front surface of the wafer.
(16) Preferably, the camera detects whether or not the tape is properly pressure-bonded to the wafer, after the pressure bonding of the tape of the tape-attached annular frame to the wafer supported by the wafer table is executed.
(17) According to the processing apparatus of the present invention, the adhesive tape that has an adhesive layer laid on a sheet or the thermocompression bonding tape that does not have an adhesive layer on a sheet can selectively be used as the tape to be pressure-bonded to the annular frame. In addition, work of integrating the wafer and the annular frame through the tape can automatically be executed even when the wafer has the ring-shaped reinforcing part in a protrusion shape formed in a region on the back surface corresponding to the outer circumferential surplus region.
(18) The above and other objects, features and advantages of the present invention and the manner of realizing them will become more apparent, and the invention itself will best be understood from a study of the following description and appended claims with reference to the attached drawings showing a preferred embodiment of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a processing apparatus of an embodiment of the present invention;
(2) FIG. **2**A is a perspective view of a wafer with a reinforcing part for which processing is executed by the processing apparatus illustrated in FIG. **1**;
(3) FIG. **2**B is a perspective view of a wafer without the reinforcing part for which processing is executed by the processing apparatus illustrated in FIG. **1**;
(4) FIG. **3** is a perspective view of a wafer cassette table and so forth illustrated in FIG. **1**;
(5) FIG. **4** is a perspective view of a hand illustrated in FIG. **1**;
(6) FIG. **5** is a perspective view of a frame housing unit and so forth illustrated in FIG. **1**;
(7) FIG. **6** is a perspective view of a frame table, a first tape pressure bonding unit, and so forth illustrated in FIG. **1**;
(8) FIG. **7** is a schematic diagram of the first tape pressure bonding unit illustrated in FIG. **1**;
(9) FIG. **8** is a schematic diagram illustrating the state in which a first pressure bonding roller is moved to a pressure bonding position and a tape is pressure-bonded to one end of an annular frame;
(10) FIG. **9** is a schematic diagram illustrating the state in which the first pressure bonding roller has been moved from the state illustrated in FIG. **8**;
(11) FIG. **10** is a schematic diagram illustrating the state in which the first pressure bonding roller has been further moved from the state illustrated in FIG. **9**;
(12) FIG. **11** is an exploded perspective view of a second tape pressure bonding unit illustrated in FIG. **1**;
(13) FIG. **12**A is a perspective view when the state in which a back surface of the wafer with the reinforcing part illustrated in FIG. **2**A is pressure-bonded to the tape of a tape-attached annular frame is viewed from the lower side;
(14) FIG. **12**B is a perspective view when the state in which a front surface of the wafer with the reinforcing part illustrated in FIG. **2**A is pressure-bonded to the tape of the tape-attached annular frame is viewed from the lower side;
(15) FIG. **13**A is a perspective view when the state in which a back surface of the wafer without the reinforcing part illustrated in FIG. **2**B is pressure-bonded to the tape of the tape-attached annular

frame is viewed from the lower side;
(16) FIG. **13**B is a perspective view when the state in which a front surface of the wafer without the reinforcing part illustrated in FIG. **2**B is pressure-bonded to the tape of the tape-attached annular frame is viewed from the lower side;
(17) FIG. **14** is a perspective view of a reinforcing part removing unit illustrated in FIG. **1**;
(18) FIG. **15** is a schematic diagram illustrating the state in which the root of a ring-shaped reinforcing part of the wafer is being irradiated with a laser beam in a reinforcing part removal step;
(19) FIG. **16** is a perspective view of a first raising-lowering table of the reinforcing part removing unit illustrated in FIG. **1**;
(20) FIG. **17**A is a perspective view of a separating part of the reinforcing part removing unit illustrated in FIG. **1**;
(21) FIG. **17**B is an enlarged perspective view of support boards illustrated in FIG. **17**A;
(22) FIG. **18** is a perspective view of a discard part of the reinforcing part removing unit illustrated in FIG. **1**;
(23) FIG. **19** is a schematic diagram illustrating the state in which tops are brought into contact with a table head illustrated in FIG. **1** and the outer diameter of the table head is detected;
(24) FIG. **20** is a schematic diagram illustrating the state in which the wafer is held under suction by a second raising-lowering table in the reinforcing part removal step;
(25) FIG. **21** is a schematic diagram illustrating the state in which the tops of the reinforcing part removing unit are caused to act on the outer circumference of the ring-shaped reinforcing part in the reinforcing part removal step;
(26) FIG. **22** is a schematic diagram illustrating the state in which the reinforcing part has been separated from the wafer in the reinforcing part removal step;
(27) FIG. **23** is a perspective view of an inversion mechanism of ring-free unit carrying-out means illustrated in FIG. **1**;
(28) FIG. **24** is a perspective view of a ring-free unit support part and a pushing part of the ring-free unit carrying-out means illustrated in FIG. **1**; and
(29) FIG. **25** is a perspective view illustrating the state in which a ring-free unit housing step is being executed.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(30) A processing apparatus of an embodiment of the present invention will be described below with reference to the drawings.
(31) (Processing Apparatus **2**)
(32) Referring to FIG. **1**, a processing apparatus, which is denoted by numeral **2** in the entire description, includes a wafer cassette table **8** on which a wafer cassette **6** having multiple wafers housed therein is placed, a wafer carrying-out unit **10** that carries a wafer out from the wafer cassette **6** placed on the wafer cassette table **8**, and a wafer table **12** that supports the wafer carried out by the wafer carrying-out unit **10**.
(33) (Wafer **4**)
(34) In FIG. **2**A, a wafer **4** for which processing is executed by the processing apparatus **2** is illustrated. On a front surface **4***a* of the wafer **4** illustrated in FIG. **2**A, a device region **18** and an outer circumferential surplus region **20** that surrounds the device region **18** are formed. The device region **18** is a region in which multiple devices **14** such as ICs and LSI circuits are formed in respective areas demarcated by planned dividing lines **16** in a lattice manner. In FIG. **2**A, a boundary **22** between the device region **18** and the outer circumferential surplus region **20** is illustrated by a two-dot chain line for convenience. However, the line that indicates the boundary **22** does not exist in practice. Further, an ID **23** for identifying the wafer **4** is given to the front surface **4***a* of the wafer **4**. The ID **23** can be configured in a form of a barcode, for example. A ring-shaped reinforcing part **24** is formed into a protrusion shape in a region on a back surface **4***b* of the

wafer **4** corresponding to the outer circumferential surplus region **20**, and the thickness of the outer circumferential surplus region **20** is thus larger than that of the device region **18**. A notch **26** that indicates the crystal orientation is formed at the circumferential edge of the wafer **4**.
(35) Moreover, the wafer for which processing is executed by the processing apparatus **2** may be a wafer **4**′ in which the ring-shaped reinforcing part is not made on a back surface **4***b*′ as illustrated in FIG. **2**B. In the following explanation, description will be made mainly regarding the case in which the wafer **4** having the reinforcing part **24** is processed.
(36) (Wafer Cassette **6** and Wafer Cassette Table **8**)
(37) As illustrated in FIG. **3**, in the wafer cassette **6**, multiple wafers **4** are housed at intervals in the upward-downward direction in the state in which the front surfaces **4***a* or the back surfaces **4***b* are oriented upward. The wafer cassette table **8** of the present embodiment has a top plate **28** on which the wafer cassette **6** is placed, and a support plate **30** that supports the top plate **28**. The top plate **28** may be capable of rising and lowering, and raising-lowering means that raises and lowers the top plate **28** and positions it to a desired height may be disposed.
(38) (Wafer Carrying-Out Unit **10**)
(39) The description will be continued with reference to FIG. **3**. The wafer carrying-out unit **10** includes a Y-axis movable member **32** that can move in a Y-axis direction indicated by an arrow Y in FIG. **3** and a Y-axis feed mechanism **34** that moves the Y-axis movable member **32** in the Y-axis direction. The Y-axis feed mechanism **34** has a ball screw **36** that is coupled to the lower end of the Y-axis movable member **32** and that extends in the Y-axis direction and a motor **38** that rotates the ball screw **36**. The Y-axis feed mechanism **34** converts rotational motion of the motor **38** to linear motion by the ball screw **36** and transmits the linear motion to the Y-axis movable member **32** to move the Y-axis movable member **32** in the Y-axis direction along a pair of guide rails **40** extending in the Y-axis direction.
(40) An X-axis direction indicated by an arrow X in FIG. **3** is a direction orthogonal to the Y-axis direction, and a Z-axis direction indicated by an arrow Z in FIG. **3** is the upward-downward direction orthogonal to the X-axis direction and the Y-axis direction. An XY plane defined by the X-axis direction and the Y-axis direction is substantially horizontal.
(41) As illustrated in FIG. **3**, the wafer carrying-out unit **10** of the present embodiment includes a conveying arm **42** and a hand **44** that is disposed at the tip of the conveying arm **42**. The hand **44** supports the front surface **4***a* or the back surface **4***b* of the wafer **4** housed in the wafer cassette **6** and inverts the front and back sides of the wafer **4**. The conveying arm **42** is disposed on the upper surface of the Y-axis movable member **32** and is driven by an appropriate drive source (not illustrated) such as an air drive source or an electric drive source. This drive source drives the conveying arm **42** and moves the hand **44** to a desired position in each of the X-axis direction, the Y-axis direction, and the Z-axis direction. In addition, the drive source inverts the hand **44** upside down.
(42) Referring to FIG. **4**, it is preferable that the hand **44** be a Bernoulli pad that generates a negative pressure by a jet of air to support the wafer **4** in a contactless manner. The hand **44** of the present embodiment has a C-shape as a whole, and multiple air jet ports **46** connected to a compressed air supply source (not illustrated) are formed in a single surface of the hand **44**. Multiple guide pins **48** are annexed to the outer circumferential edge of the hand **44** at intervals in the circumferential direction. Each guide pin **48** is configured to be movable in the radial direction of the hand **44**.
(43) As illustrated in FIG. **3** and FIG. **4**, after positioning the hand **44** to the lower surface side of the wafer **4** in the wafer cassette **6** placed on the wafer cassette table **8**, the wafer carrying-out unit **10** jets compressed air from the air jet ports **46** of the hand **44** to generate a negative pressure on the lower surface side of the hand **44** by the Bernoulli effect, and supports the wafer **4** under suction from the lower surface side in a contactless manner by the hand **44**. Horizontal movement of the wafer **4** supported under suction by the hand **44** is restricted by the respective guide pins **48**.

Further, the wafer carrying-out unit **10** carries the wafer **4** supported under suction by the hand **44** out from the wafer cassette **6** by moving the Y-axis movable member **32** and the conveying arm **42**.
(44) (Notch Detecting Unit **50** of Wafer Carrying-Out Unit **10**)
(45) As illustrated in FIG. **4**, the wafer carrying-out unit **10** of the present embodiment includes a notch detecting unit **50** that detects the position of the notch **26** of the wafer **4**. The notch detecting unit **50** may include, for example, a light emitting element **52** and a light receiving element **54** that are spaced from each other in the upward-downward direction as well as a drive source (not illustrated) that rotates at least one of the guide pins **48** of the hand **44**.
(46) The light emitting element **52** and the light receiving element **54** can be annexed to the Y-axis movable member **32** or a conveyance route with the interposition of an appropriate bracket (not illustrated). Further, when the guide pin **48** rotates by the above-described drive source, the wafer **4** supported under suction by the hand **44** rotates together with the rotation of the guide pin **48**. It is preferable that the outer circumferential surface of the guide pin **48** that rotates by the drive source be formed of appropriate synthetic rubber in order to surely transmit the rotation from the guide pin **48** to the wafer **4**.
(47) The notch detecting unit **50** can detect the position of the notch **26** by rotating the wafer **4** by the drive source through the guide pin **48** in the state in which the wafer **4** is supported under suction by the hand **44** and in which the outer circumference of the wafer **4** is positioned between the light emitting element **52** and the light receiving element **54**. This makes it possible to adjust the orientation of the wafer **4** to a desired orientation.
(48) (Wafer Table **12**)
(49) As illustrated in FIG. **3**, the wafer table **12** is disposed adjacent to the wafer carrying-out unit **10**. The wafer table **12** of the present embodiment includes a wafer support part **56** that supports the wafer **4** and a frame support part **58** that is disposed around the outer circumference of the wafer support part **56** and that supports an annular frame **64** (see FIG. **5**) to be described later. Multiple suction holes **60** are formed in the upper surface of the wafer support part **56**, and each suction hole **60** is connected to suction means (not illustrated).
(50) When the hand **44** is inverted by 180° to invert the front and back sides of the wafer **4** and the wafer **4** is then placed on the wafer table **12**, the wafer **4** is supported by the wafer support part **56**. Further, after supporting the wafer **4** by the wafer support part **56**, the wafer table **12** holds the wafer **4** under suction through actuating the suction means and generating a suction force for each suction hole **60**.
(51) When the wafer **4** is supported by the wafer table **12**, the wafer **4** may be in either the state in which the front surface **4***a* of the wafer **4** is oriented downward or the state in which the back surface **4***b* of the wafer **4** is oriented downward. However, when the wafer **4** is supported by the wafer table **12** in the state in which the back surface **4***b* of the wafer **4** having the ring-shaped reinforcing part **24** thereon is oriented downward, an annular recess **62** that can house the ring-shaped reinforcing part **24** therein is made in the upper surface of the wafer table **12**. In such a case, when the wafer **4** is placed on the wafer table **12**, the reinforcing part **24** is housed in the annular recess **62**, and a portion that is positioned inside the reinforcing part **24** on the back surface **4***b* of the wafer **4** comes into contact with the upper surface of the wafer table **12**.
(52) Referring to FIG. **5**, the processing apparatus **2** further includes a frame housing unit **66** that houses multiple annular frames **64** therein, each annular frame **64** having an opening part **64***a* for housing the wafer **4** therein, a frame carrying-out unit **68** that carries an annular frame **64** out from the frame housing unit **66**, and a frame table **70** that supports the annular frame **64** carried out by the frame carrying-out unit **68**.
(53) (Frame Housing Unit **66**)
(54) As illustrated in FIG. **5**, the frame housing unit **66** of the present embodiment includes a housing **72**, a rising-lowering plate **74** disposed to be capable of rising and lowering in the housing **72**, and raising-lowering means (not illustrated) that raises and lowers the rising-lowering plate **74**.

(55) A Z-axis guide member **78** extending in the Z-axis direction is disposed on the side surface of the housing **72** on the far side in the X-axis direction in FIG. **5**. The rising-lowering plate **74** is supported by the Z-axis guide member **78** in such a manner as to be capable of rising and lowering, and the raising-lowering means that raises and lowers the rising-lowering plate **74** is disposed inside the Z-axis guide member **78**. The raising-lowering means may include, for example, a ball screw that is coupled to the rising-lowering plate **74** and that extends in the Z-axis direction and a motor that rotates this ball screw.
(56) A door **76** to which a handle **76***a* is annexed is disposed at the side surface of the housing **72** on the near side in the X-axis direction in FIG. **5**. In the frame housing unit **66**, the annular frames **64** can be housed inside the housing **72** when the handle **76***a* is used to open the door **76**. Further, an opening part **80** is made at the upper end of the housing **72**.
(57) As illustrated in FIG. **5**, the annular frames **64** are housed in the housing **72** in such a manner as to be stacked on top of one another over the upper surface of the rising-lowering plate **74**. The annular frame **64** on the top of the multiple stacked annular frames **64** is carried out from the opening part **80** of the housing **72** by the frame carrying-out unit **68**. Further, when the annular frame **64** is carried out from the opening part **80**, the frame housing unit **66** raises the rising-lowering plate **74** as appropriate by the raising-lowering means and moves the top annular frame **64** to the position from which this annular frame **64** can be carried out by the frame carrying-out unit **68**.
(58) (Frame Carrying-Out Unit **68**)
(59) The description will be continued with reference to FIG. **5**. The frame carrying-out unit **68** includes an X-axis guide member **82** that is fixed to an appropriate bracket (not illustrated) and that extends in the X-axis direction, an X-axis movable member **84** supported by the X-axis guide member **82** movably in the X-axis direction, an X-axis feed mechanism (not illustrated) that moves the X-axis movable member **84** in the X-axis direction, a Z-axis movable member **86** supported by the X-axis movable member **84** movably in the Z-axis direction, and a Z-axis feed mechanism (not illustrated) that moves the Z-axis movable member **86** in the Z-axis direction.
(60) The X-axis feed mechanism of the frame carrying-out unit **68** may include a ball screw that is coupled to the X-axis movable member **84** and that extends in the X-axis direction and a motor that rotates this ball screw. The Z-axis feed mechanism may include a ball screw that is coupled to the Z-axis movable member **86** and that extends in the Z-axis direction and a motor that rotates this ball screw.
(61) The Z-axis movable member **86** of the frame carrying-out unit **68** has a holding part **88** that holds the annular frame **64**. The holding part **88** of the present embodiment has a rectangular board **90** and multiple suction pads **92** disposed on a lower surface of the board **90**. Each suction pad **92** is connected to suction means (not illustrated).
(62) The frame carrying-out unit **68** holds under suction the top annular frame **64** housed in the frame housing unit **66**, by the suction pads **92** of the holding part **88**, and then moves the X-axis movable member **84** and the Z-axis movable member **86**. Thus, the frame carrying-out unit **68** carries the top annular frame **64**, which is held under suction, out from the frame housing unit **66**.
(63) (Frame Table **70**)
(64) As illustrated in FIG. **5**, the frame table **70** is supported by a Z-axis guide member **94** in such a manner as to be capable of rising and lowering. An appropriate drive source (for example, air drive source or electric drive source) that raises and lowers the frame table **70** is annexed to the Z-axis guide member **94**.
(65) As illustrated in FIG. **1** and FIG. **5**, the processing apparatus **2** includes a first tape pressure bonding unit **98** (see FIG. **1**) that is disposed above the frame table **70** and that includes a first pressure bonding roller **110** for executing pressure bonding of a tape **96** to the annular frame **64**, and a tape-attached frame conveying unit **100** (see FIG. **5**) that conveys the annular frame **64** to which the tape **96** is pressure-bonded (hereinafter, often referred to as a “tape-attached annular

frame **64**′″) to the wafer table **12** and that places the tape-attached annular frame **64**′ on the wafer table **12** in such a manner that the front surface **4***a* or the back surface **4***b* of the wafer **4** supported by the wafer table **12** is positioned in the opening part **64***a* of the annular frame **64**. The processing apparatus **2** further includes a second tape pressure bonding unit **102** (see FIG. **1**) that includes a second pressure bonding roller **174** for executing pressure bonding of the tape **96** of the tape-attached annular frame **64**′ to the front surface **4***a* or the back surface **4***b* of the wafer **4**.
(66) (First Tape Pressure Bonding Unit **98**)
(67) Referring to FIG. **6**, the first tape pressure bonding unit **98** of the present embodiment includes a roll tape support part **104** that supports a roll tape **96**R where the tape **96** that is yet to be used is wound, a tape take-up part **106** that takes up the tape **96** that has been used, a tape pull-out part **108** that pulls out the tape **96** from the roll tape **96**R, the first pressure bonding roller **110** that executes pressure bonding of the pulled-out tape **96** to the annular frame **64**, and a cutting part **112** that cuts the tape **96** protruding to the outer circumference of the annular frame **64** along the annular frame **64**.
(68) (Roll Tape Support Part **104** of First Tape Pressure Bonding Unit **98**)
(69) As illustrated in FIG. **6**, the roll tape support part **104** includes a support roller **114** supported by an appropriate bracket (not illustrated) rotatably around an axis line extending in the X-axis direction. The roll tape **96**R where the tape **96** having a pressure bonding surface to which release paper **116** for protecting the pressure bonding surface of the tape **96** is annexed is wound in a circular cylindrical form is supported by the support roller **114**.
(70) (Tape Take-Up Part **106** of First Tape Pressure Bonding Unit **98**)
(71) The tape take-up part **106** includes a take-up roller **118** supported by an appropriate bracket (not illustrated) rotatably around an axis line extending in the X-axis direction and a motor (not illustrated) that rotates the take-up roller **118**. As illustrated in FIG. **6**, the tape take-up part **106** takes up the used tape **96** in which a circular opening part **120** corresponding to a portion stuck to the annular frame **64** is formed, by rotating the take-up roller **118** by the motor.
(72) (Tape Pull-Out Part **108** of First Tape Pressure Bonding Unit **98**)
(73) The description will be continued with reference to FIG. **6**. The tape pull-out part **108** includes a pull-out roller **122** disposed below the support roller **114** of the roll tape support part **104**, a motor (not illustrated) that rotates the pull-out roller **122**, and a driven roller **124** that rotates together with the rotation of the pull-out roller **122**. The tape pull-out part **108** pulls out the tape **96** sandwiched between the pull-out roller **122** and the driven roller **124** from the roll tape **96**R by rotating the driven roller **124** together with the pull-out roller **122** by the motor.
(74) The release paper **116** is separated from the tape **96** that has passed between the pull-out roller **122** and the driven roller **124**, and the separated release paper **116** is taken up by a release paper take-up part **126**. The release paper take-up part **126** of the present embodiment has a release paper take-up roller **128** disposed above the driven roller **124** and a motor (not illustrated) that rotates the release paper take-up roller **128**. Further, the tape **96** from which the release paper **116** has been separated goes through a pair of guide rollers **130** spaced from the pull-out roller **122** in the Y-axis direction and is guided to the take-up roller **118**.
(75) (First Pressure Bonding Roller **110** of First Tape Pressure Bonding Unit **98**)
(76) Referring to FIG. **7** and FIG. **8**, the first pressure bonding roller **110** is raised and lowered by a Z-axis feed mechanism (not illustrated) and is moved to an upper-side standby position illustrated in FIG. **7** and a lower-side pressure bonding position illustrated in FIG. **8**. Moreover, the first pressure bonding roller **110** is moved in the Y-axis direction by a Y-axis feed mechanism, which is not illustrated in the diagram, in the state of being at the lower-side pressure bonding position. The Y-axis feed mechanism and the Z-axis feed mechanism can include an appropriate drive source (for example, air drive source or electric drive source).
(77) (Cutting Part **112** of First Tape Pressure Bonding Unit **98**)
(78) As illustrated in FIG. **6**, the cutting part **112** includes a Z-axis guide member **134** that is fixed

to an appropriate bracket (not illustrated) and that extends in the Z-axis direction, a Z-axis movable member **136** supported by the Z-axis guide member **134** movably in the Z-axis direction, and a Z-axis feed mechanism (not illustrated) that moves the Z-axis movable member **136** in the Z-axis direction. The Z-axis feed mechanism of the cutting part **112** may include a ball screw that is coupled to the Z-axis movable member **136** and that extends in the Z-axis direction and a motor that rotates this ball screw.
(79) Further, the cutting part **112** includes a motor **138** fixed to the lower surface of the tip of the Z-axis movable member **136** and an arm piece **140** rotated by the motor **138** around an axis line extending in the Z-axis direction. To the lower surface of the arm piece **140**, first and second drooping pieces **142***a* and **142***b* spaced from each other are annexed. A circular cutter **144** is supported by the first drooping piece **142***a* rotatably around an axis line orthogonal to the Z-axis direction. A holding-down roller **146** is supported by the second drooping piece **142***b* rotatably around an axis line orthogonal to the Z-axis direction.
(80) (Tape **96**)
(81) The tape **96** may be an adhesive tape that has an adhesive layer (glue layer) laid on a single surface of a sheet, or may be a thermocompression bonding tape that does not have an adhesive layer laid on a sheet. The thermocompression bonding tape is a tape of a thermoplastic synthetic resin (for example, polyolefin-based resin) and is a tape that softens or melts to exert an adhesive force when being heated to a temperature near the melting point.
(82) The pressure bonding surface (adhesive surface) of the adhesive tape is given the release paper **116**, whereas the pressure bonding surface of the thermocompression bonding tape is given the release paper **116** in some cases and is not given the release paper **116** in other cases. When the thermocompression bonding tape that is not given the release paper **116** is used, the first tape pressure bonding unit **98** does not need to include the release paper take-up part **126** because there is no need to collect the release paper **116**.
(83) In the case in which the tape **96** is the thermocompression bonding tape, a first heating unit (not illustrated) is disposed in one of or both the frame table **70** and the first pressure bonding roller **110**, and one of or both the frame table **70** and the first pressure bonding roller **110** are heated to a temperature near the melting point of the thermocompression bonding tape when pressure bonding of the thermocompression bonding tape to the annular frame **64** is executed.
(84) When pressure bonding of the tape **96** to the annular frame **64** is executed by the first tape pressure bonding unit **98**, before the annular frame **64** is placed on the frame table **70**, the frame table **70** is first lowered to a position at which the frame table **70** can receive the annular frame **64** (for example, position illustrated in FIG. **6**). In addition, the first pressure bonding roller **110** is moved to the upper-side standby position (for example, position illustrated in FIG. **6**).
(85) Further, the tape **96** is pulled out from the roll tape **96**R, and the tape **96** from which the release paper **116** has been separated is positioned above the frame table **70** in the state in which the tape **96** is stretched tight without being slacked. When the tape **96** is an adhesive tape, the pressure bonding surface (adhesive surface) of the tape **96** located above the frame table **70** is oriented downward.
(86) Subsequently, when the annular frame **64** is carried out by the frame carrying-out unit **68** and placed on the frame table **70**, the frame table **70** is moved to a pressure bonding start position (position illustrated in FIG. **8**). In addition, the first pressure bonding roller **110** is moved to the lower-side pressure bonding position (position illustrated in FIG. **8**), and tension is applied to the tape **96**, so that pressure bonding of the tape **96** to one end of the annular frame **64** is executed.
(87) Next, the first pressure bonding roller **110** is moved in the Y-axis direction toward the other end of the annular frame **64** while the tape **96** is pressed against the annular frame **64** by the first pressure bonding roller **110**. Thus, pressure bonding of the tape **96** to the annular frame **64** is executed in the state in which uniform tension is applied to the tape **96**.
(88) After the tape **96** is pressure-bonded to the annular frame **64**, the first tape pressure bonding

unit **98** lowers the Z-axis movable member **136** of the cutting part **112** by the Z-axis feed mechanism to press the cutter **144** against the tape **96** on the annular frame **64** and hold down the annular frame **64** by the holding-down roller **146** from over the tape **96**.
(89) Subsequently, the first tape pressure bonding unit **98** rotates the arm piece **140** by the motor **138** and causes the cutter **144** and the holding-down roller **146** to move to draw a circle along the annular frame **64**. This can cut the tape **96** protruding to the outer circumference of the annular frame **64** along the annular frame **64**.
(90) Further, because the annular frame **64** is held down by the holding-down roller **146** from over the tape **96**, deviation of the positions of the annular frame **64** and the tape **96** is prevented when the tape **96** is being cut. Then, after the frame table **70** is lowered, the used tape **96** in which the circular opening part **120** corresponding to a portion stuck to the annular frame **64** is formed is taken up by the tape take-up part **106**.
(91) (Tape-Attached Frame Conveying Unit **100**)
(92) As illustrated in FIG. **5**, the tape-attached frame conveying unit **100** includes a Y-axis guide member **148** that is fixed to an appropriate bracket (not illustrated) and that extends in the Y-axis direction, a Y-axis movable member **150** supported by the Y-axis guide member **148** movably in the Y-axis direction, a Y-axis feed mechanism (not illustrated) that moves the Y-axis movable member **150** in the Y-axis direction, a Z-axis movable member **152** supported by the Y-axis movable member **150** movably in the Z-axis direction, and a Z-axis feed mechanism (not illustrated) that moves the Z-axis movable member **152** in the Z-axis direction.
(93) The Y-axis feed mechanism of the tape-attached frame conveying unit **100** may include a ball screw that is coupled to the Y-axis movable member **150** and that extends in the Y-axis direction and a motor that rotates this ball screw. The Z-axis feed mechanism may include a ball screw that is coupled to the Z-axis movable member **152** and that extends in the Z-axis direction and a motor that rotates this ball screw.
(94) The Z-axis movable member **152** of the tape-attached frame conveying unit **100** has a holding part **154** that holds the tape-attached annular frame **64′**. The holding part **154** of the present embodiment has a rectangular board **156** and multiple suction pads **158** disposed on the lower surface of the board **156**. Each suction pad **158** is connected to suction means (not illustrated).
(95) The tape-attached frame conveying unit **100** holds under suction the upper surface of the tape-attached annular frame **64′** supported by the frame table **70**, by the respective suction pads **158** of the holding part **154**, and moves the Y-axis movable member **150** and the Z-axis movable member **152**. Thus, the tape-attached frame conveying unit **100** conveys the tape-attached annular frame **64′** held under suction by the holding part **154** from the frame table **70** to the wafer table **12**, and places the tape-attached annular frame **64′** on the wafer table **12** in such a manner that the front surface **4***a* or the back surface **4***b* of the wafer **4** supported by the wafer table **12** is positioned in the opening part **64***a* of the annular frame **64**.
(96) (Second Tape Pressure Bonding Unit **102**)
(97) As illustrated in FIG. **11**, the second tape pressure bonding unit **102** includes an upper chamber **160** disposed above the wafer table **12**, a lower chamber **162** in which the wafer table **12** is housed, and a raising-lowering mechanism **164** that raises and lowers the upper chamber **160** and that generates a closed state in which the upper chamber **160** is brought into contact with the lower chamber **162** and an opened state in which the upper chamber **160** is separated from the lower chamber **162**. The second tape pressure bonding unit **102** further includes a vacuum part **166** that sets the upper chamber **160** and the lower chamber **162** to a vacuum state in the closed state and an opening-to-atmosphere part **168** that opens the upper chamber **160** and the lower chamber **162** to the atmosphere.
(98) (Upper Chamber **160** and Raising-Lowering Mechanism **164** of Second Tape Pressure Bonding Unit **102**)
(99) As illustrated in FIG. **11**, the upper chamber **160** of the present embodiment includes a circular

top plate **170** and a circular cylindrical sidewall **172** that droops from the circumferential edge of the top plate **170**. The raising-lowering mechanism **164** that can include an appropriate actuator such as an air cylinder is mounted on the upper surface of the top plate **170**. In a housing space defined by the lower surface of the top plate **170** and the inner circumferential surface of the sidewall **172**, the second pressure bonding roller **174** for pressure bonding of the tape **96** of the tape-attached annular frame **64′** to the front surface **4*a*** or the back surface **4*b*** of the wafer **4** supported by the wafer table **12**, a support piece **176** that rotatably supports the second pressure bonding roller **174**, and a Y-axis feed mechanism **178** that moves the support piece **176** in the Y-axis direction are disposed.

(100) The Y-axis feed mechanism **178** has a ball screw **180** that is coupled to the support piece **176** and that extends in the Y-axis direction and a motor **182** that rotates the ball screw **180**. Further, the Y-axis feed mechanism **178** converts rotational motion of the motor **182** to linear motion by the ball screw **180** and transmits the linear motion to the support piece **176** to move the support piece **176** along a pair of guide rails **184** extending in the Y-axis direction.

(101) As illustrated in FIG. **11**, a camera **185** that images the wafer **4** supported by the wafer table **12** is disposed in the upper chamber **160**. The camera **185** images the wafer **4** on the wafer table **12** and detects whether the exposed surface (upper surface) of the wafer **4** is the front surface **4*a*** or the back surface **4*b***. Further, the camera **185** acquires the ID **23** displayed on the front surface **4*a*** of the wafer **4**, when the exposed surface (upper surface) of the wafer **4** supported by the wafer table **12** is the front surface **4*a***. Moreover, after the tape **96** of the tape-attached annular frame **64′** is pressure-bonded to the wafer **4** supported by the wafer table **12**, the camera **185** detects whether or not the tape **96** is properly pressure-bonded to the wafer **4**.

(102) (Lower Chamber **162**, Vacuum Part **166**, and Opening-to-Atmosphere Part **168** of Second Tape Pressure Bonding Unit **102**)

(103) As illustrated in FIG. **11**, the lower chamber **162** has a circular cylindrical sidewall **186**. The upper part of the sidewall **186** is opened, and the lower part of the sidewall **186** is closed. A connection opening **188** is formed in the sidewall **186**. The vacuum part **166** that can include an appropriate vacuum pump is connected to the connection opening **188** through a flow path **190**. The opening-to-atmosphere part **168** that can include an appropriate valve capable of opening the flow path **190** to the atmosphere is disposed on the flow path **190**.

(104) In the state in which the tape **96** of the tape-attached annular frame **64′** is positioned to the front surface **4*a*** or the back surface **4*b*** of the wafer **4** supported by the wafer table **12**, the second tape pressure bonding unit **102** lowers the upper chamber **160** by the raising-lowering mechanism **164** and brings the lower end of the sidewall **172** of the upper chamber **160** into contact with the upper end of the sidewall **186** of the lower chamber **162** to set the upper chamber **160** and the lower chamber **162** to the closed state. In addition, the second tape pressure bonding unit **102** brings the second pressure bonding roller **174** into contact with the tape-attached annular frame **64′**.

(105) Subsequently, the second tape pressure bonding unit **102** actuates the vacuum pump that configures the vacuum part **166**, in the state in which the valve that configures the opening-to-atmosphere part **168** is closed, to set the inside of the upper chamber **160** and the lower chamber **162** to a vacuum state. Thereafter, the second tape pressure bonding unit **102** rolls the second pressure bonding roller **174** in the Y-axis direction by the Y-axis feed mechanism **178**, to thereby execute pressure bonding of the tape **96** to the front surface **4*a*** or the back surface **4*b*** of the wafer **4** and generate a frame unit U.

(106) When the tape **96** is pressure-bonded to the back surface **4*b*** of the wafer **4** having the reinforcing part **24**, a slight gap is formed between the wafer **4** and the tape **96** at the root of the ring-shaped reinforcing part **24**. However, because the pressure bonding of the wafer **4** and the tape **96** is executed in the state in which the inside of the upper chamber **160** and the lower chamber **162** is set to the vacuum state, the pressure of the slight gap between the wafer **4** and the tape **96** is lower than the atmospheric pressure. Thus, when the opening-to-atmosphere part **168** is opened

after the tape **96** is pressure-bonded, the tape **96** is pressed against the wafer **4** by the atmospheric pressure. With this, the gap between the wafer **4** and the tape **96** at the root of the reinforcing part **24** disappears, and the tape **96** comes into close contact with the back surface **4***b* of the wafer **4** along the root of the reinforcing part **24**.

(107) In the case in which the tape **96** is the thermocompression bonding tape, a second heating unit (not illustrated) is disposed in one of or both the wafer table **12** and the second pressure bonding roller **174**, and one of or both the wafer table **12** and the second pressure bonding roller **174** are heated to a temperature near the melting point of the thermocompression bonding tape when pressure bonding of the tape **96** of the tape-attached annular frame **64′** to the wafer **4** is executed.

(108) As illustrated in FIG. **1** and FIG. **14**, the processing apparatus **2** of the present embodiment further includes frame unit carrying-out means **192** that carries, out from the wafer table **12**, the frame unit U in which the tape **96** of the tape-attached annular frame **64′** and the front surface **4***a* or the back surface **4***b* of the wafer **4** are pressure-bonded together by the second tape pressure bonding unit **102**, and a reinforcing part removing unit **194** that cuts and removes the ring-shaped reinforcing part **24** from the wafer **4** of the frame unit U carried out by the frame unit carrying-out means **192**. The processing apparatus **2** also includes ring-free unit carrying-out means **196** (see FIG. **1**) that carries a ring-free unit resulting from the removal of the ring-shaped reinforcing part **24** out from the reinforcing part removing unit **194**, and a frame cassette table **200** (see FIG. **1**) on which a frame cassette **198** that houses the ring-free unit carried out by the ring-free unit carrying-out means **196** is placed.

(109) (Frame Unit Carrying-Out Means **192**)

(110) As illustrated in FIG. **14**, the frame unit carrying-out means **192** of the present embodiment includes a frame unit holding part **202** including a wafer holding part **202***a* that holds the wafer **4** and a frame holding part **202***b* that holds the annular frame **64**, and a conveying part **206** that conveys the frame unit holding part **202** to a temporary placement table **204**.

(111) (Frame Unit Holding Part **202** of Frame Unit Carrying-Out Means **192**)

(112) The wafer holding part **202***a* of the frame unit holding part **202** includes a circular board **208** and a circular suction adhesion piece **210** mounted on the lower surface of the board **208**. Multiple suction holes (not illustrated) are formed in the lower surface of the suction adhesion piece **210**, and each suction hole is connected to suction means (not illustrated). The frame holding part **202***b* includes multiple (in the present embodiment, four) protruding pieces **212** that protrude outward in the radial direction from the circumferential edge of the board **208** of the wafer holding part **202***a* and that are arranged at intervals in the circumferential direction, and suction pads **214** annexed to the lower surfaces of the protruding pieces **212**. Each suction pad **214** is connected to the suction means (not illustrated).

(113) (Conveying Part **206** of Frame Unit Carrying-Out Means **192**)

(114) The conveying part **206** includes an X-axis guide member **216** that is fixed to an appropriate bracket (not illustrated) and that extends in the X-axis direction, an X-axis movable member **218** supported by the X-axis guide member **216** movably in the X-axis direction, and an X-axis feed mechanism (not illustrated) that moves the X-axis movable member **218** in the X-axis direction. The conveying part **206** further includes a Z-axis movable member **220** supported by the X-axis movable member **218** movably in the Z-axis direction, a Z-axis feed mechanism (not illustrated) that moves the Z-axis movable member **220** in the Z-axis direction, a Y-axis movable member **222** supported by the Z-axis movable member **220** movably in the Y-axis direction, and a Y-axis feed mechanism (not illustrated) that moves the Y-axis movable member **222** in the Y-axis direction. The board **208** of the wafer holding part **202***a* is coupled to the tip of the Y-axis movable member **222**. Each of the X-axis, Y-axis, and Z-axis feed mechanisms of the conveying part **206** may include a ball screw and a motor that rotates the ball screw.

(115) It is preferable for the frame unit carrying-out means **192** to include a two-dimensional

movement mechanism that two-dimensionally moves the frame unit holding part **202** in the horizontal direction and an imaging part **224** that images the outer circumference of the wafer **4** of the frame unit U held by the frame unit holding part **202**. In the present embodiment, the frame unit holding part **202** two-dimensionally moves in the horizontal direction in the XY plane by the X-axis feed mechanism and the Y-axis feed mechanism of the conveying part **206**, and the two-dimensional movement mechanism is configured by the conveying part **206**. Further, the imaging part **224** of the present embodiment is disposed between the wafer table **12** and the temporary placement table **204** and images the outer circumference of the wafer **4** of the frame unit U held by the frame unit holding part **202** from the lower side of the wafer **4**.
(116) The frame unit carrying-out means **192** carries the frame unit U held by the frame unit holding part **202** out from the wafer table **12** by actuating the conveying part **206** in the state in which the wafer **4** is held under suction by the suction adhesion piece **210** of the wafer holding part **202***a* and in which the annular frame **64** is held under suction by the suction pads **214** of the frame holding part **202***b*.
(117) Further, the frame unit carrying-out means **192** of the present embodiment measures the coordinates of at least three points at the outer circumference of the wafer **4** by actuating the conveying part **206**, which configures the two-dimensional movement mechanism, and imaging, by the imaging part **224**, at least three places at the outer circumference of the wafer **4** of the frame unit U held by the frame unit holding part **202**, and obtains the center coordinates of the wafer **4** on the basis of the measured coordinates of the three points. Then, the frame unit carrying-out means **192** makes the center of the wafer **4** correspond with the center of the temporary placement table **204** and temporarily places the frame unit U on the temporary placement table **204**.
(118) (Temporary Placement Table **204**)
(119) As illustrated in FIG. **14**, the temporary placement table **204** is spaced from the wafer table **12** in the X-axis direction. It is preferable that the temporary placement table **204** include a heater (not illustrated), that the tape **96** of the frame unit U temporarily placed on the temporary placement table **204** be softened by being heated by the heater, and that the tape **96** be brought into closer contact with the root of the ring-shaped reinforcing part **24** by the atmospheric pressure.
(120) (Temporary Placement Table Conveying Part **232**)
(121) The processing apparatus **2** of the present embodiment includes a temporary placement table conveying part **232** that conveys the temporary placement table **204** in the Y-axis direction. The temporary placement table conveying part **232** includes a Y-axis guide member **234** extending in the Y-axis direction, a Y-axis movable member **236** supported by the Y-axis guide member **234** movably in the Y-axis direction, and a Y-axis feed mechanism **238** that moves the Y-axis movable member **236** in the Y-axis direction. The temporary placement table **204** is fixed to the upper part of the Y-axis movable member **236**. The Y-axis feed mechanism **238** has a ball screw **240** that is coupled to the Y-axis movable member **236** and that extends in the Y-axis direction and a motor **242** that rotates the ball screw **240**. Further, the temporary placement table conveying part **232** converts rotational motion of the motor **242** to linear motion by the ball screw **240** and transmits the linear motion to the Y-axis movable member **236** to convey the temporary placement table **204** in the Y-axis direction together with the Y-axis movable member **236**.
(122) (Reinforcing Part Removing Unit **194**)
(123) As illustrated in FIG. **1** and FIG. **14**, the reinforcing part removing unit **194** includes a laser beam irradiation unit **244** that irradiates the root of the ring-shaped reinforcing part **24**, which is formed at the outer circumference of the wafer **4**, with a laser beam to form a cut groove, a first raising-lowering table **246** (see FIG. **1**) that holds and raises the frame unit U temporarily placed on the temporary placement table **204** and that moves the frame unit U in the X-axis direction to position it to the laser beam irradiation unit **244**, and a separating part **248** that separates the ring-shaped reinforcing part **24** from the cut groove.
(124) (Laser Beam Irradiation Unit **244** of Reinforcing Part Removing Unit **194**)

(125) As illustrated in FIG. **14**, the laser beam irradiation unit **244** includes a housing **250** disposed adjacent to the temporary placement table **204** in the X-axis direction, a laser oscillator (not illustrated) that is housed in the housing **250** and that oscillates a laser, and a light collector **252** that focuses the laser beam emitted by the laser oscillator and that irradiates, with the laser beam, the root of the ring-shaped reinforcing part **24** formed at the outer circumference of the wafer **4**. The laser beam irradiation unit **244** further includes a suction nozzle **254** that sucks debris generated when the wafer **4** is irradiated with the laser beam, and suction means (not illustrated) connected to the suction nozzle **254**.
(126) The light collector **252** extends upward from the upper surface of the housing **250** with an inclination toward the side of the suction nozzle **254**. With this, dropping of debris generated in the irradiation with the laser beam onto the light collector **252** is suppressed. Further, the suction nozzle **254** extends upward from the upper surface of the housing **250** with an inclination toward the side of the light collector **252**.
(127) As illustrated in FIG. **15**, the laser beam irradiation unit **244** irradiates the root of the ring-shaped reinforcing part **24**, which is formed at the outer circumference of the wafer **4**, with a laser beam LB while the frame unit U held by the first raising-lowering table **246** is rotated, to form a ring-shaped cut groove **256** along the root of the reinforcing part **24** by ablation processing. Moreover, the laser beam irradiation unit **244** sucks debris generated due to the ablation processing by the suction nozzle **254**.
(128) (First Raising-Lowering Table **246** of Reinforcing Part Removing Unit **194**)
(129) As illustrated in FIG. **1**, the first raising-lowering table **246** is disposed above the temporary placement table **204** movably in the X-axis direction and movably in the Z-axis direction. Referring to FIG. **16**, the first raising-lowering table **246** includes an X-axis guide member **258** that is fixed to an appropriate bracket (not illustrated) and that extends in the X-axis direction, an X-axis movable member **260** supported by the X-axis guide member **258** movably in the X-axis direction, an X-axis feed mechanism (not illustrated) that moves the X-axis movable member **260** in the X-axis direction, a Z-axis movable member **262** supported by the X-axis movable member **260** movably in the Z-axis direction, and a Z-axis feed mechanism (not illustrated) that moves the Z-axis movable member **262** in the Z-axis direction. Each of the X-axis and Z-axis feed mechanisms of the first raising-lowering table **246** may include a ball screw and a motor that rotates the ball screw.
(130) A support shaft **264** extending downward is rotatably supported by the lower surface of the tip of the Z-axis movable member **262**. A motor **266** that rotates the support shaft **264** around an axis line extending in the Z-axis direction is attached to the upper surface of the tip of the Z-axis movable member **262**. A circular suction adhesion piece **268** is fixed to the lower end of the support shaft **264**. In the lower surface of the suction adhesion piece **268**, multiple suction holes (not illustrated) are formed on a circumference corresponding to the size of the annular frame **64** at intervals in the circumferential direction. Each suction hole is connected to suction means.
(131) The first raising-lowering table **246** holds the annular frame **64** of the frame unit U under suction by the suction adhesion piece **268**. Thereafter, the first raising-lowering table **246** moves the Z-axis movable member **262** and the X-axis movable member **260** to raise the frame unit U held under suction by the suction adhesion piece **268** and move the frame unit U in the X-axis direction, positioning it to the laser beam irradiation unit **244**. When the annular frame **64** is formed of a material having magnetism, an electromagnet (not illustrated) may be annexed to the lower surface of the suction adhesion piece **268**, and the suction adhesion piece **268** may cause adhesion of the annular frame **64** by a magnetic force.
(132) Further, when the wafer **4** is irradiated with the laser beam LB by the laser beam irradiation unit **244**, the first raising-lowering table **246** actuates the motor **266** to rotate the frame unit U held under suction by the suction adhesion piece **268**. Moreover, the first raising-lowering table **246** moves the frame unit U in which the cut groove **256** has been formed at the root of the reinforcing

part **24**, in the X-axis direction and the Z-axis direction, and temporarily places it on the temporary placement table **204**.
(133) (Separating Part **248** of Reinforcing Part Removing Unit **194**)
(134) As illustrated in FIG. **1**, the separating part **248** is spaced from the first raising-lowering table **246** in the Y-axis direction in the movable range of the temporary placement table **204** in the Y-axis direction. Referring to FIG. **17**A, FIG. **17**B, and FIG. **18**, the separating part **248** includes ultraviolet irradiation parts **270** (see FIG. **17**A) that irradiate a portion of the tape **96** corresponding to the cut groove **256** with ultraviolet to reduce the adhesive force of the tape **96**, and a second raising-lowering table **272** (see FIG. **17**A) that holds an inner portion of the wafer **4** under suction in such a manner that the ring-shaped reinforcing part **24** is exposed from the outer circumference. The separating part **248** further includes a separator **274** (see FIG. **17**A) that causes tops **402** having a wedge to act on the outer circumference of the ring-shaped reinforcing part **24** and separates the ring-shaped reinforcing part **24**, and a discard part **276** (see FIG. **18**) by which the separated ring-shaped reinforcing part **24** is discarded.
(135) As illustrated in FIG. **17**A, the separating part **248** of the present embodiment includes a Z-axis guide member **278** that is fixed to an appropriate bracket (not illustrated) and that extends in the Z-axis direction, a Z-axis movable member **280** supported by the Z-axis guide member **278** movably in the Z-axis direction, and raising-lowering means (not illustrated) that moves the Z-axis movable member **280** in the Z-axis direction. The raising-lowering means may include a ball screw that is coupled to the Z-axis movable member **280** and that extends in the Z-axis direction and a motor that rotates this ball screw.
(136) On the lower surface of the tip of the Z-axis movable member **280**, a support piece **282** is supported, and the second raising-lowering table **272** is rotatably supported. A motor **284** that rotates the second raising-lowering table **272** is attached to the upper surface of the tip of the Z-axis movable member **280**. A pair of the above-described ultraviolet irradiation parts **270** spaced from each other in the Y-axis direction are annexed to the support piece **282** of the present embodiment.
(137) The second raising-lowering table **272** has a support shaft **286** extending downward from the lower surface of the tip of the Z-axis movable member **280** and a circular table head **287** attachably-detachably mounted on the lower end of the support shaft **286**. Multiple suction holes (not illustrated) are formed in the lower surface of the table head **287**, and each suction hole is connected to suction means.
(138) The table head **287** has an outer diameter corresponding to the inner diameter of the reinforcing part **24** of the wafer **4**. Specifically, the diameter of the table head **287** is slightly smaller than that of the device region **18** of the wafer **4**. Further, the table head **287** is attachably-detachably mounted on the support shaft **286** and can be replaced according to the diameter of the wafer **4**. The support shaft **286** on which the table head **287** is mounted is connected to the raising-lowering means of the separating part **248** through the Z-axis movable member **280**. As described above, the second raising-lowering table **272** includes two or more kinds of table heads **287** having an outer diameter corresponding to the inner diameter of the reinforcing part **24** of the wafer **4**, and the table head **287** is attachably-detachably mounted to the raising-lowering means of the separating part **248**.
(139) Further, the above-described separator **274** is mounted on the support piece **282**. The separator **274** includes a pair of movable pieces **288** spaced from each other and disposed on the lower surface of the support piece **282** movably in the longitudinal direction of the support piece **282**, a pair of feed means **290** that move the pair of movable pieces **288**, a pair of support boards **400** supported by the respective movable pieces **288** in such a manner as to be capable of rising and lowering, and a pair of Z-axis feed mechanisms **294** that raise and lower the pair of support boards **400** in the Z-axis direction. The pair of feed means **290** and the pair of Z-axis feed mechanisms **294** each can include an appropriate actuator such as an air cylinder or an electric cylinder.
(140) The description will be continued with reference to FIG. **17**A and FIG. **17**B. On the upper

surface of the support board **400**, the tops **402** having a wedge, frame support parts **404** that support the annular frame **64**, and an ionizer **406** that removes static electricity from the frame unit U are mounted.
(141) The top **402** has an inverted truncated cone shape in which the diameter gradually decreases from the upper side toward the lower side, and the wedge is formed by an upper surface **402***a* and a side surface **402***b* of the top **402**. A pair of tops **402** are disposed on the upper surface of each support board **400** and spaced from each other. Also, the pair of tops **402** are supported by the support board **400** rotatably around an axis line extending in the Z-axis direction.
(142) A pair of frame support parts **404** are disposed adjacent to the tops **402** on the upper surface of each of the support boards **400**. The frame support parts **404** each have a housing **404***a* fixed to the support board **400** and a sphere **404***b* rotatably supported by the housing **404***a*. In the frame support parts **404**, the annular frame **64** is supported by the respective spheres **404***b*.
(143) The ionizer **406** is disposed adjacent to the tops **402**. The ionizer **406** removes static electricity from the frame unit U by blowing ionized air toward the frame unit U.
(144) The separating part **248** of the present embodiment includes detecting means (not illustrated) that detects whether or not the kind of table head **287** input to a control unit (not illustrated) that controls the operation of the processing apparatus **2** corresponds with the kind of table head **287** actually mounted in the processing apparatus **2**.
(145) The control unit include a computer having a central processing unit (CPU) that executes calculation processing according to a control program, a read only memory (ROM) that stores the control program and so forth, and a readable-writable random access memory (RAM) that stores a calculation result and so forth. Processing conditions such as the diameter of the wafer **4**, the width of the reinforcing part **24**, and the outer diameter of the table head **287** are input to the control unit by an operator.
(146) The detecting means of the present embodiment includes the tops **402** of the separator **274** and the feed means **290** that cause the tops **402** to get closer to and get further away from the table head **287** by actuating the movable pieces **288**. Further, in the detecting means, before processing of the wafer **4** is started, the movable pieces **288** are actuated by the feed means **290**, and, as illustrated in FIG. **19**, whether or not the outer diameter of the table head **287** obtained through bringing the tops **402** of the separator **274** into contact with the outer circumference of the table head **287** corresponds with the outer diameter of the table head **287** input to the control unit is detected. When it is detected by the detecting means that both do not correspond with each other, an error notification (for example, display indicating the non-correspondence on a control panel (not illustrated)) is made.
(147) Even when the wafers **4** have the same diameter of, for example, 200 mm, the width of the ring-shaped reinforcing part **24** may be different among the wafers **4**, for example, 3 mm, 5 mm, and so forth, in some cases. Thus, the table head **287** corresponding to the device region **18** of the wafer **4** needs to be mounted in the processing apparatus **2**. If the kind of table head **287** input to the control unit does not correspond with the kind of table head **287** actually mounted, it becomes impossible to properly remove the ring-shaped reinforcing part **24** from the wafer **4**.
(148) Regarding this point, the processing apparatus **2** of the present embodiment includes the detecting means that detects whether or not the kind of table head **287** input to the control unit corresponds with the kind of table head **287** actually mounted in the processing apparatus **2**. Therefore, whether or not the proper table head **287** corresponding to the wafer **4** is mounted can be checked before processing of the wafer **4** is started, and the ring-shaped reinforcing part **24** can properly be removed from the wafer **4** at the time of the processing of the wafer **4**.
(149) Referring to FIG. **18**, the discard part **276** includes a belt conveyor **300** that conveys the separated ring-shaped reinforcing part **24** and a dust box **302** in which the ring-shaped reinforcing part **24** conveyed by the belt conveyor **300** is housed. The belt conveyor **300** is moved by an appropriate actuator (not illustrated) to a collection position (position illustrated by solid lines in

FIG. **18**) at which the belt conveyor **300** substantially horizontally extends and a standby position (position illustrated by two-dot chain lines in FIG. **18**) at which the belt conveyor **300** substantially vertically extends.

(150) A door **304** to which a handle **304***a* is annexed is disposed at the side surface of the dust box **302** on the near side in the X-axis direction in FIG. **18**. A crusher (not illustrated) that crushes the collected ring-shaped reinforcing part **24** is attached inside the dust box **302**. The dust box **302** is configured in such a manner that crushed waste of the ring-shaped reinforcing part **24** housed in the dust box **302** can be taken out when the handle **304***a* is used to open the door **304**.

(151) When the temporary placement table **204** on which the frame unit U in which the cut groove **256** has been formed at the root of the reinforcing part **24** is temporarily placed is positioned below the separating part **248** by the temporary placement table conveying part **232**, as illustrated in FIG. **20**, the separating part **248** holds the inner portion of the wafer **4** under suction by the second raising-lowering table **272** with the ring-shaped reinforcing part **24** exposed from the outer circumference. Subsequently, the movable pieces **288** are moved by the feed means **290**, and the support boards **400** are moved by the Z-axis feed mechanisms **294**. Then, as illustrated in FIG. **21**, the tops **402** having the wedge are caused to act on the outer circumference of the ring-shaped reinforcing part **24**. Specifically, the wedges of the tops **402** are positioned between the tape **96** and the reinforcing part **24**. Further, the lower surface of the annular frame **64** is brought into contact with the spheres **404***b* of the frame support parts **404**, and the annular frame **64** is supported by the spheres **404***b*.

(152) Next, irradiation with ultraviolet from the pair of ultraviolet irradiation parts **270** is executed to reduce the adhesive force of the tape **96** sticking to the ring-shaped reinforcing part **24**. In addition, the frame unit U is rotated together with the second raising-lowering table **272** by the motor **284** relative to the separator **274**. Thus, the tape **96** in which the adhesive force has been reduced is detached from the reinforcing part **24** by the wedges of the tops **402**. Thus, as illustrated in FIG. **22**, the ring-shaped reinforcing part **24** can be separated from the frame unit U. The separated reinforcing part **24** is conveyed to the dust box **302** by the belt conveyor **300** and is collected. When the reinforcing part **24** is separated, the separator **274** may be rotated relative to the frame unit U.

(153) Moreover, when the reinforcing part **24** is separated, ionized air is blown from the ionizer **406** toward the frame unit U. With this, even when static electricity is generated due to contact of the tops **402** with the tape **96** and the reinforcing part **24**, the static electricity is removed by the ionized air blown from the ionizer **406**. This keeps the tape **96** and the reinforcing part **24** from being attracted to each other due to the static electricity, and the reinforcing part **24** is surely separated from the frame unit U.

(154) When the reinforcing part **24** is separated, together with the relative rotation of the frame unit U and the separator **274**, the tops **402** that act on the frame unit U rotate, and the spheres **404***b* in contact with the lower surface of the annular frame **64** also rotate. Therefore, the relative rotation of the frame unit U and the separator **274** is smoothly executed.

(155) (Ring-Free Unit Carrying-Out Means **196**)

(156) As illustrated in FIG. **1**, the ring-free unit carrying-out means **196** is disposed adjacent to the reinforcing part removing unit **194**. Referring to FIG. **23** and FIG. **24**, the ring-free unit carrying-out means **196** of the present embodiment includes an inversion mechanism **308** (see FIG. **23**) that includes a frame holding part **306** facing the ring-free unit supported by the second raising-lowering table **272** and being configured to hold the annular frame **64** and that moves toward the frame cassette table **200** and also inverts the frame holding part **306**, a ring-free unit support part **310** (see FIG. **24**) that supports the ring-free unit inverted by the inversion mechanism **308**, and a pushing part **312** (see FIG. **24**) that causes the ring-free unit supported by the ring-free unit support part **310** to enter the frame cassette **198** placed on the frame cassette table **200** and be housed therein.

(157) (Inversion Mechanism **308** of Ring-Free Unit Carrying-Out Means **196**)
(158) As illustrated in FIG. **23**, the inversion mechanism **308** includes a Y-axis guide member **314** extending in the Y-axis direction, a Y-axis movable member **316** supported by the Y-axis guide member **314** movably in the Y-axis direction, a Y-axis feed mechanism (not illustrated) that moves the Y-axis movable member **316** in the Y-axis direction, an arm **318** supported by the Y-axis movable member **316** movably in the Z-axis direction, and a Z-axis feed mechanism (not illustrated) that moves the arm **318** in the Z-axis direction. Each of the Y-axis and Z-axis feed mechanisms of the inversion mechanism **308** may include a ball screw and a motor that rotates the ball screw.
(159) The above-described frame holding part **306** is supported by the arm **318** in such a manner as to be capable of vertically inverting. In addition, a motor **320** that vertically inverts the frame holding part **306** is attached to the arm **318**. The frame holding part **306** of the present embodiment includes a board **324** rotatably supported by the arm **318** through a pair of rotating shafts **322** and multiple suction pads **326** annexed to a single surface of the board **324**. Each suction pad **326** is connected to suction means (not illustrated). Further, one rotating shaft **322** is coupled to the motor **320**.
(160) In the state in which the suction pads **326** are oriented upward, the inversion mechanism **308** holds under suction, by the suction pads **326**, the lower surface of the annular frame **64** of a ring-free unit U′ supported by the second raising-lowering table **272** and receives the ring-free unit U′ from the second raising-lowering table **272**. Further, the inversion mechanism **308** inverts the frame holding part **306** by the motor **320** and then moves the ring-free unit U′ held by the frame holding part **306** toward the frame cassette table **200** by moving the Y-axis movable member **316**.
(161) (Ring-Free Unit Support Part **310** of Ring-Free Unit Carrying-Out Means **196**)
(162) As illustrated in FIG. **24**, the ring-free unit support part **310** of the present embodiment includes a pair of support plates **328** supported through an appropriate bracket (not illustrated) movably in the X-axis direction and space adjusting means (not illustrated) that adjusts the space between the pair of support plates **328** in the X-axis direction. The space adjusting means can include an appropriate actuator such as an air cylinder or an electric cylinder.
(163) A heater (not illustrated) is mounted on the pair of support plates **328** that support the ring-free unit U′. In the state in which the space between the pair of support plates **328** is narrowed, the pair of support plates **328** heat the tape **96** of the ring-free unit U′ by the heater and thus eliminate slack and wrinkles of the tape **96** caused due to the removal of the reinforcing part **24**.
(164) (Pushing Part **312** of Ring-Free Unit Carrying-Out Means **196**)
(165) The description will be continued with reference to FIG. **24**. The pushing part **312** of the present embodiment includes a Y-axis guide member **330** extending in the Y-axis direction, a Y-axis movable member **332** supported by the Y-axis guide member **330** movably in the Y-axis direction, and a Y-axis feed mechanism (not illustrated) that moves the Y-axis movable member **332** in the Y-axis direction. The Y-axis movable member **332** has a base part **334** supported by the Y-axis guide member **330**, a support column **336** extending upward from the upper surface of the base part **334**, and a pressing piece **338** annexed to the upper end of the support column **336**. The Y-axis feed mechanism of the pushing part **312** may include a ball screw that is coupled to the Y-axis movable member **332** and that extends in the Y-axis direction and a motor that rotates this ball screw.
(166) As illustrated in FIG. **25**, the ring-free unit support part **310** expands the space between the pair of support plates **328** by the space adjusting means before receiving the ring-free unit U′, and then receives the ring-free unit U′ held by the suction pads **326**. When the ring-free unit support part **310** has received the ring-free unit U′, the pushing part **312** then moves the Y-axis movable member **332** in the Y-axis direction by the Y-axis feed mechanism and thus causes, with the use of the pressing piece **338**, the ring-free unit U′ supported by the ring-free unit support part **310** to enter the frame cassette **198** placed on the frame cassette table **200**, so that the ring-free unit U′ is housed in the frame cassette **198**.

(167) (Frame Cassette **198** and Frame Cassette Table **200**)

(168) In the frame cassette **198** illustrated in FIG. **1** and FIG. **25**, multiple ring-free units U′ (frame units U in the case of wafers **4**′ without the reinforcing part) are housed at intervals in the upward-downward direction. As illustrated in FIG. **24** and FIG. **25**, the frame cassette table **200** includes a placement part **340** on which the frame cassette **198** is placed and a raising-lowering part **342** that raises and lowers the placement part **340** and that positions it to a desired height. The raising-lowering part **342** may include a ball screw that is coupled to the placement part **340** and that extends in the Z-axis direction and a motor that rotates this ball screw.

(169) Next, description will be made regarding a processing method of, by using the above-described processing apparatus **2**, executing pressure bonding of the tape **96** to the wafer **4**, which has the ring-shaped reinforcing part **24** in a protrusion shape formed in a region on the back surface **4***b* corresponding to the outer circumferential surplus region **20**, to integrate the wafer **4** with the annular frame **64** and cutting and removing the ring-shaped reinforcing part **24** from the wafer **4**.

(170) (Wafer Cassette Placement Step)

(171) In the present embodiment, as illustrated in FIG. **1** and FIG. **3**, a wafer cassette placement step is first executed to place the wafer cassette **6** having multiple wafers **4** housed therein on the wafer cassette table **8**. In the wafer cassette **6** of the present embodiment, multiple wafers **4** are housed at intervals in the upward-downward direction.

(172) (Frame Housing Step)

(173) Further, as illustrated in FIG. **1** and FIG. **5**, a frame housing step is executed to house multiple ring-shaped annular frames **64** each having the opening part **64***a* for housing the wafer **4** therein in the frame housing unit **66**. The frame housing step may be executed before the wafer cassette placement step or may be executed after the wafer cassette placement step.

(174) In the frame housing step, after the rising-lowering plate **74** of the frame housing unit **66** is lowered to a desired position, the handle **76***a* is used to open the door **76**, and the multiple annular frames **64** are stacked on top of one another over the upper surface of the rising-lowering plate **74**, so that the multiple annular frames **64** are housed in the frame housing unit **66**. Further, the height of the rising-lowering plate **74** is adjusted as appropriate, and the top annular frame **64** is moved to a position from which this annular frame **64** can be carried out by the frame carrying-out unit **68**.

(175) (Wafer Carrying-Out Step)

(176) After the wafer cassette placement step and the frame housing step are executed, a wafer carrying-out step is executed to carry the wafer **4** out from the wafer cassette **6** placed on the wafer cassette table **8**.

(177) Referring to FIG. **3**, in the wafer carrying-out step, the Y-axis feed mechanism **34** of the wafer carrying-out unit **10** is first actuated, and the Y-axis movable member **32** is positioned near the wafer cassette table **8**. Subsequently, the conveying arm **42** is driven, and the hand **44** in which the air jet ports **46** are oriented upward is positioned to the lower surface side of the wafer **4** in the wafer cassette **6**. When the hand **44** is positioned to the lower surface side of the wafer **4**, a gap is made between the lower surface of the wafer **4** and the hand **44**, and each guide pin **48** is positioned outside in the radial direction.

(178) Next, compressed air is jetted from the air jet ports **46** of the hand **44** to generate a negative pressure on the lower surface side of the hand **44** by the Bernoulli effect, and the wafer **4** is supported under suction from the lower surface side by the hand **44** in a contactless manner. Subsequently, each guide pin **48** is moved inward in the radial direction, and thus, horizontal movement of the wafer **4** supported under suction by the hand **44** is restricted by the respective guide pins **48**. Then, the Y-axis movable member **32** and the conveying arm **42** of the wafer carrying-out unit **10** are moved, and the wafer **4** supported under suction by the hand **44** is carried out from the wafer cassette **6**.

(179) (Notch Detection Step)

(180) It is preferable to execute a notch detection step of detecting the position of the notch **26** of

the wafer **4** after executing the wafer carrying-out step. In the notch detection step, as illustrated in FIG. **4**, the outer circumference of the wafer **4** supported under suction by the hand **44** is positioned between the light emitting element **52** and the light receiving element **54** of the notch detecting unit **50**. Next, when the wafer **4** is rotated by the drive source through the guide pin **48**, the position of the notch **26** of the wafer **4** is detected. This makes it possible to adjust the orientation of the wafer **4** to a desired orientation.
(181) (Wafer Support Step)
(182) After the notch detection step is executed, a wafer support step is executed to support, by the wafer table **12**, the wafer **4** carried out by the wafer carrying-out unit **10**.
(183) Referring to FIG. **3**, in the wafer support step, the hand **44** of the wafer carrying-out unit **10** is first inverted upside down. Subsequently, the Y-axis movable member **32** and the conveying arm **42** of the wafer carrying-out unit **10** are moved, and the wafer **4** supported under suction by the hand **44** is brought into contact with the wafer support part **56** of the wafer table **12**. At this time, the wafer **4** may be in either the state in which the front surface **4***a* of the wafer **4** is oriented downward or the state in which the back surface **4***b* of the wafer **4** is oriented downward. When the back surface **4***b* having the ring-shaped reinforcing part **24** formed thereon is oriented downward, the ring-shaped reinforcing part **24** is housed in the annular recess **62** of the wafer table **12**.
(184) Next, the wafer **4** is held under suction by actuating the suction means of the wafer table **12** and generating a suction force for each suction hole **60**. Subsequently, the suction support of the wafer **4** by the hand **44** is released, and the hand **44** is separated from the wafer table **12**. In this manner, the wafer **4** is transferred from the wafer carrying-out unit **10** to the wafer table **12**. Because the wafer **4** transferred to the wafer table **12** is held under suction by each suction hole **60**, the position of the wafer **4** does not deviate.
(185) After the wafer **4** is transferred to the wafer table **12**, the wafer **4** on the wafer table **12** is imaged from the upper side by the camera **185** (see FIG. **11**) disposed in the upper chamber **160**. This can detect whether the exposed surface (upper surface) of the wafer **4** is the front surface **4***a* or the back surface **4***b*, and check whether or not the surface that should be set as the exposed surface is oriented upward. If the surface that should be set as the exposed surface of the wafer **4** is not oriented upward, information indicating such a wrong orientation is displayed on the control panel (not illustrated). Further, when the exposed surface of the wafer **4** is the front surface **4***a*, the ID **23** displayed on the front surface **4***a* of the wafer **4** can be acquired from an image captured by the camera **185**.
(186) (Frame Carrying-Out Step)
(187) Further, after the wafer cassette placement step and the frame housing step are executed, concurrently with the wafer carrying-out step and the wafer support step, a frame carrying-out step is executed to carry the annular frame **64** out from the frame housing unit **66**.
(188) Referring to FIG. **5**, in the frame carrying-out step, the X-axis movable member **84** and the Z-axis movable member **86** of the frame carrying-out unit **68** are first moved, and the suction pads **92** of the holding part **88** are brought into contact with the upper surface of the top annular frame **64** housed in the frame housing unit **66**. Next, the top annular frame **64** is held under suction by the suction pads **92** by actuating the suction means of the frame carrying-out unit **68** and generating a suction force for the suction pads **92**. Then, the X-axis movable member **84** and the Z-axis movable member **86** of the frame carrying-out unit **68** are moved, and the top annular frame **64** held under suction by the suction pads **92** of the holding part **88** is carried out from the frame housing unit **66**.
(189) (Frame Support Step)
(190) After the frame carrying-out step is executed, a frame support step is executed to support, by the frame table **70**, the annular frame **64** carried out by the frame carrying-out unit **68**.
(191) The description will be continued with reference to FIG. **5**. In the frame support step, the X-axis movable member **84** and the Z-axis movable member **86** of the frame carrying-out unit **68** are first moved, and the annular frame **64** held under suction by the suction pads **92** is brought into

contact with the upper surface of the frame table **70**. At this time, the frame table **70** is moved in advance to a position at which the frame table **70** can receive the annular frame **64**. Subsequently, the suction force of the suction pads **92** of the frame carrying-out unit **68** is deactivated to place the annular frame **64** on the frame table **70**. Then, the X-axis movable member **84** and the Z-axis movable member **86** of the frame carrying-out unit **68** are moved, and the holding part **88** is separated from the upper side of the frame table **70**.
(192) (First Tape Pressure Bonding Step)
(193) After the frame support step is executed, a first tape pressure bonding step is performed to execute pressure bonding of the tape **96** to the annular frame **64**.
(194) Referring to FIG. **6** and FIG. **7**, in the first tape pressure bonding step, the tape **96** is first pulled out from the roll tape **96**R, and the tape **96** from which the release paper **116** has been separated is positioned above the frame table **70** in the state in which the tape **96** is stretched tight without being slacked. When the tape **96** is an adhesive tape, the pressure bonding surface of the tape **96** located above the frame table **70** is oriented downward.
(195) Next, the frame table **70** is raised to the pressure bonding start position (position illustrated in FIG. **8**), and the first pressure bonding roller **110** is lowered to the lower-side pressure bonding position (position illustrated in FIG. **8**). Thus, tension is applied to the tape **96**, and pressure bonding of the tape **96** to one end of the annular frame **64** is executed.
(196) When the frame table **70** is moved to the pressure bonding start position and the first pressure bonding roller **110** is moved to the lower-side pressure bonding position as described above, as illustrated in FIG. **8**, an elevation angle α is formed between the annular frame **64** and the tape **96** extending from the first pressure bonding roller **110** toward the guide roller **130** on the lower side.
(197) Next, the first pressure bonding roller **110** is moved in the Y-axis direction toward the other end of the annular frame **64** while the tape **96** is pressed against the annular frame **64** by the first pressure bonding roller **110**. Thus, pressure bonding of the tape **96** to the annular frame **64** can be executed in the state in which uniform tension is applied to the tape **96**.
(198) When the first pressure bonding roller **110** is moved, it is preferable to keep the tension applied to the tape **96** constant by raising the frame table **70** in synchronization with the movement of the first pressure bonding roller **110**. Specifically, as illustrated in FIG. **9** and FIG. **10**, the frame table **70** is gradually raised in synchronization with the movement of the first pressure bonding roller **110** in such a manner that the elevation angle α is constant.
(199) When the tape **96** is a thermocompression bonding tape, the temperature of the upper surface of the frame table **70** or the temperature of the outer circumferential surface of the first pressure bonding roller **110** is adjusted to a temperature at which the tape **96** softens or melts, and then, the first pressure bonding roller **110** is rolled in the Y-axis direction while constant tension is applied to the tape **96** by the first pressure bonding roller **110**. Thus, thermocompression bonding of the tape **96** to the annular frame **64** can be executed. Needless to say, also when the thermocompression bonding of the tape **96** is executed, the frame table **70** is gradually raised in synchronization with the movement of the first pressure bonding roller **110** in such a manner that the elevation angle α is constant.
(200) Subsequently, the cutter **144** and the holding-down roller **146** of the cutting part **112** of the first tape pressure bonding unit **98** are lowered to press the cutter **144** against the tape **96** on the annular frame **64** and hold down the annular frame **64** by the holding-down roller **146** from over the tape **96**. Next, the arm piece **140** is rotated by the motor **138**, and the cutter **144** and the holding-down roller **146** are moved to draw a circle along the annular frame **64**. Thus, the tape **96** protruding to the outer circumference of the annular frame **64** can be cut along the annular frame **64**.
(201) Further, because the annular frame **64** is held down by the holding-down roller **146** from over the tape **96**, deviation of the positions of the annular frame **64** and the tape **96** is prevented when the tape **96** is being cut. The used tape **96** in which the circular opening part **120** is formed is taken

up by the tape take-up part **106**.
(202) (Tape-Attached Frame Conveyance Step)
(203) After the first tape pressure bonding step is executed, a tape-attached frame conveyance step is executed to convey the annular frame **64** to which the tape **96** is pressure-bonded, i.e., the tape-attached annular frame **64**′, to the wafer table **12** and place the tape-attached annular frame **64**′ on the wafer table **12** in such a manner that the front surface **4***a* or the back surface **4***b* of the wafer **4** supported by the wafer table **12** is positioned in the opening part **64***a* of the annular frame **64**.
(204) In the tape-attached frame conveyance step, the frame table **70** is first lowered to a position from which the tape-attached annular frame **64**′ can be carried out by the tape-attached frame conveying unit **100**. Subsequently, the Y-axis movable member **150** and the Z-axis movable member **152** of the tape-attached frame conveying unit **100** (see FIG. **5**) are moved, and the respective suction pads **158** of the holding part **154** of the tape-attached frame conveying unit **100** are brought into contact with the upper surface of the tape-attached annular frame **64**′ (see FIG. **11**) supported by the frame table **70**.
(205) Next, the upper surface of the tape-attached annular frame **64**′ is held under suction by the suction pads **158** by actuating the suction means of the tape-attached frame conveying unit **100** and generating a suction force for the suction pads **158**. Subsequently, the Y-axis movable member **150** and the Z-axis movable member **152** of the tape-attached frame conveying unit **100** are moved, and the tape-attached annular frame **64**′ held under suction by the suction pads **158** is carried out from the frame table **70**.
(206) Next, the tape-attached annular frame **64**′ held under suction by the suction pads **158** of the tape-attached frame conveying unit **100** is conveyed to the wafer table **12**. Then, as illustrated in FIG. **11**, the opening part **64***a* of the annular frame **64** is positioned to the wafer **4** supported by the wafer table **12**, and the tape-attached annular frame **64**′ is brought into contact with the frame support part **58** of the wafer table **12**. At this time, the pressure bonding surface of the tape **96** of the tape-attached annular frame **64**′ is oriented downward. Further, in the present embodiment, the back surface **4***b* of the wafer **4** is oriented upward and is facing the pressure bonding surface of the tape **96** as illustrated in FIG. **11**.
(207) Subsequently, the suction force of the suction pads **158** of the tape-attached frame conveying unit **100** is deactivated to place the tape-attached annular frame **64**′ on the frame support part **58** of the wafer table **12**. Then, the Y-axis movable member **150** and the Z-axis movable member **152** of the tape-attached frame conveying unit **100** are moved, and the holding part **154** is separated from the upper side of the wafer table **12**.
(208) (Second Tape Pressure Bonding Step)
(209) After the tape-attached frame conveyance step is executed, a second tape pressure bonding step is performed to execute pressure bonding of the tape **96** of the tape-attached annular frame **64**′ to the front surface **4***a* or the back surface **4***b* of the wafer **4**.
(210) Referring to FIG. **11**, in the second tape pressure bonding step, the upper chamber **160** is first lowered by the raising-lowering mechanism **164** of the second tape pressure bonding unit **102**, and the lower end of the sidewall **172** of the upper chamber **160** is brought into contact with the upper end of the sidewall **186** of the lower chamber **162**. Thus, the upper chamber **160** and the lower chamber **162** are set to the closed state. In addition, the second pressure bonding roller **174** is brought into contact with the tape-attached annular frame **64**′. Thereupon, the upper end of the ring-shaped reinforcing part **24** of the wafer **4** sticks to the pressure bonding surface of the tape **96** of the tape-attached annular frame **64**′.
(211) Next, the vacuum part **166** of the second tape pressure bonding unit **102** is actuated in the state in which the opening-to-atmosphere part **168** is closed, and the inside of the upper chamber **160** and the lower chamber **162** is set to a vacuum state. Subsequently, pressure bonding of the tape **96** to the wafer **4** is executed by rolling the second pressure bonding roller **174** of the second tape pressure bonding unit **102** in the Y-axis direction. Thus, the frame unit U in which the wafer **4** and

the tape **96** are pressure-bonded together can be generated. Next, the opening-to-atmosphere part **168** is opened, and the tape **96** is brought into close contact with the wafer **4** along the root of the ring-shaped reinforcing part **24** by the atmospheric pressure. Then, the upper chamber **160** is raised by the raising-lowering mechanism **164**.
(212) When the tape **96** is a thermocompression bonding tape, the temperature of the upper surface of the wafer table **12** or the temperature of the outer circumferential surface of the second pressure bonding roller **174** is adjusted to a temperature at which the tape **96** softens or melts, and then, the second pressure bonding roller **174** is rolled. Thus, thermocompression bonding of the tape **96** to the wafer **4** can be executed.
(213) After the frame unit U is generated by executing the pressure bonding of the tape **96** of the tape-attached annular frame **64′** to the wafer **4**, it is preferable to image the frame unit U by the camera **185** disposed in the upper chamber **160** and detect whether or not the tape **96** is properly pressure-bonded to the wafer **4**. When the wafer **4** (for example, root of the reinforcing part **24**) is not in close contact with the tape **96** and a gap exists between the wafer **4** and the tape **96**, the processing by the processing apparatus **2** is suspended. This can suppress the occurrence of a defect of the device **14** in dicing the wafer **4** into device chips each including the device **14** in a later step.
(214) In the present embodiment, as illustrated in FIG. **12**A, the tape **96** of the tape-attached annular frame **64′** is pressure-bonded to the back surface **4***b* of the wafer **4** having the ring-shaped reinforcing part **24**. However, the tape **96** may be pressure-bonded to the front surface **4***a* of the wafer **4** as illustrated in FIG. **12**B.
(215) Further, as illustrated in FIG. **13**A and FIG. **13**B, also in the case of executing pressure bonding of the tape **96** of the tape-attached annular frame **64′** to the wafer **4′** without the reinforcing part, the surface to which the tape **96** of the tape-attached annular frame **64′** is pressure-bonded may be either a front surface **4***a′* or a back surface **4***b′* of the wafer **4′**.
(216) Incidentally, when the tape **96** is pressure-bonded to the front surface **4***a* or **4***a′* of the wafer **4** or **4′**, cutting processing can be executed for the wafer **4** or **4′** by a cutting blade from the side of the back surface **4***b* or **4***b′* in a later step. Thus, processing with high quality can be implemented for a wafer of a material with which it is preferable to execute processing from the back surface side in terms of the crystal structure (for example, SiC wafer).
(217) Moreover, in the case in which the tape **96** is pressure-bonded to the front surface **4***a* or **4***a′* of the wafer **4** or **4′**, there is an advantage also when laser processing is executed for the wafer **4** or **4′**. Specifically, when the wafer **4** or **4′** is irradiated with a laser beam with a wavelength having absorbability with respect to the wafer **4** or **4′** from the side of the back surface **4***b* or **4***b′* and ablation processing is executed, the devices **14** on the front surface **4***a* or **4***a′* are prevented from being contaminated due to debris. In addition, when the wafer **4** or **4′** is irradiated with a laser beam with a wavelength having transmissibility with respect to the wafer **4** or **4′** with the focal point of the laser beam positioned inside the wafer **4** or **4′** from the side of the back surface **4***b* or **4***b′* and processing is executed for the inside of the wafer **4** or **4′**, the processing can be performed without being affected by the devices **14**.
(218) When pressure bonding of the tape **96** to the wafer **4** or **4′** (particularly, front surface **4***a* or **4***a′*) is executed, it is desirable to use the thermocompression bonding tape free from the possibility of remaining of an adhesive layer.
(219) (Frame Unit Carrying-Out Step)
(220) After the second tape pressure bonding step is executed, a frame unit carrying-out step is executed to carry, out from the wafer table **12**, the frame unit U in which the tape **96** of the tape-attached annular frame **64′** and the front surface **4***a* or the back surface **4***b* of the wafer **4** are pressure-bonded together.
(221) Referring to FIG. **5**, in the frame unit carrying-out step, the conveying part **206** of the frame unit carrying-out means **192** is first actuated, and the lower surface of the suction adhesion piece **210** of the wafer holding part **202***a* of the frame unit holding part **202** is brought into contact with

the wafer **4** with the interposition of the tape **96**. In addition, the suction pads **214** of the frame holding part **202***b* are brought into contact with the annular frame **64**.
(222) Subsequently, a suction force is generated for the suction adhesion piece **210** of the wafer holding part **202***a* and the suction pads **214** of the frame holding part **202***b*, and the wafer **4** is held under suction by the suction adhesion piece **210** of the wafer holding part **202***a*. In addition, the annular frame **64** is held under suction by the suction pads **214** of the frame holding part **202***b*. Next, the suction holding of the wafer **4** by the wafer table **12** is released. Then, the conveying part **206** is actuated, and the frame unit U held by the frame unit holding part **202** is carried out from the wafer table **12**.
(223) (Temporary Placement Step)
(224) After the frame unit carrying-out step is executed, a temporary placement step is executed to make the center of the wafer **4** correspond with the center of the temporary placement table **204** and temporarily place the frame unit U on the temporary placement table **204**. However, regarding the wafer **4′** without the reinforcing part as illustrated in FIG. **2**B, the temporary placement step and a reinforcing part removal step to be described later are not executed, and the frame unit U carried out from the wafer table **12** is housed in the frame cassette **198**.
(225) Referring to FIG. **14**, in the temporary placement step, the frame unit U held by the frame unit holding part **202** is first positioned above the imaging part **224**. Subsequently, the conveying part **206** that configures the two-dimensional movement mechanism of the frame unit carrying-out means **192** is actuated, and at least three places at the outer circumference of the wafer **4** of the frame unit U held by the frame unit holding part **202** are imaged by the imaging part **224**. Thus, the coordinates of at least three points at the outer circumference of the wafer **4** are measured. Next, the center coordinates of the wafer **4** are obtained on the basis of the measured coordinates of the three points.
(226) Subsequently, the conveying part **206** is actuated to position the center of the wafer **4** to the center of the temporary placement table **204**, and the frame unit U is brought into contact with the upper surface of the temporary placement table **204**. Next, the suction holding of the wafer **4** by the wafer holding part **202***a* is released, and the suction holding of the annular frame **64** by the frame holding part **202***b* is also released, to transfer the frame unit U from the frame unit carrying-out means **192** to the temporary placement table **204**.
(227) Subsequently, the heater of the temporary placement table **204** is actuated to heat the tape **96** of the frame unit U temporarily placed on the temporary placement table **204**, by the heater. With this, the tape **96** softens and is brought into close contact with the root of the ring-shaped reinforcing part **24** of the wafer **4**.
(228) (Reinforcing Part Removal Step)
(229) After the temporary placement step is executed, a reinforcing part removal step is executed to cut and remove the ring-shaped reinforcing part **24** from the wafer **4** of the frame unit U carried out by the frame unit carrying-out means **192**.
(230) Referring to FIG. **1**, FIG. **14**, and FIG. **16**, in the reinforcing part removal step, the X-axis movable member **260** and the Z-axis movable member **262** of the first raising-lowering table **246** of the reinforcing part removing unit **194** are first moved, and the lower surface of the suction adhesion piece **268** is brought into contact with the upper surface of the annular frame **64** of the frame unit U temporarily placed on the temporary placement table **204**. Next, a suction force is generated for each suction hole of the suction adhesion piece **268** of the first raising-lowering table **246**, and the annular frame **64** of the frame unit U is held under suction.
(231) Subsequently, the X-axis movable member **260** and the Z-axis movable member **262** of the first raising-lowering table **246** are actuated, and, as illustrated in FIG. **15**, the frame unit U held under suction by the suction adhesion piece **268** is positioned above the laser beam irradiation unit **244**. Next, the focal point of the laser beam LB is positioned to the root of the ring-shaped reinforcing part **24** of the wafer **4** of the frame unit U. In FIG. **15**, the tape **96** is pressure-bonded to

the back surface **4***b* of the wafer **4** in which the reinforcing part **24** is disposed. However, the tape **96** may be pressure-bonded to the front surface **4***a* of the wafer **4**.
(232) Subsequently, while the suction adhesion piece **268** and the frame unit U are rotated by the motor **266** of the first raising-lowering table **246**, the root of the ring-shaped reinforcing part **24** of the wafer **4** is irradiated with the laser beam LB. This can execute ablation processing for the root of the ring-shaped reinforcing part **24** of the wafer **4** and form the ring-shaped cut groove **256**. Further, when the wafer **4** is irradiated with the laser beam LB, the suction means of the laser beam irradiation unit **244** is actuated to generate a suction force for the suction nozzle **254**, and debris generated due to the ablation processing is sucked by the suction nozzle **254**.
(233) Next, the X-axis movable member **260** and the Z-axis movable member **262** of the first raising-lowering table **246** are moved, and the frame unit U held under suction by the suction adhesion piece **268** is brought into contact with the upper surface of the temporary placement table **204**. Subsequently, the suction force of the suction adhesion piece **268** of the first raising-lowering table **246** is deactivated to transfer the frame unit U from the first raising-lowering table **246** to the temporary placement table **204**.
(234) Next, the temporary placement table **204** that has received the frame unit U is positioned below the separating part **248** of the reinforcing part removing unit **194** by the temporary placement table conveying part **232** (see FIG. **14**). At this time, the belt conveyor **300** of the discard part **276** is moved to the standby position. Subsequently, the second raising-lowering table **272** of the separating part **248** is lowered, and the lower surface of the second raising-lowering table **272** is brought into contact with the tape **96** on the inner portion of the wafer **4**. Next, a suction force is generated at the lower surface of the second raising-lowering table **272**, and, as illustrated in FIG. **20**, the inner portion of the wafer **4** of the frame unit U is held under suction by the table head **287** of the second raising-lowering table **272** in the state in which the ring-shaped reinforcing part **24** is exposed from the outer circumference.
(235) Subsequently, the second raising-lowering table **272** that holds the wafer **4** of the frame unit U under suction is raised to separate the frame unit U from the temporary placement table **204**. In addition, the temporary placement table **204** is moved to the lower side of the first raising-lowering table **246**. Next, the movable pieces **288** are moved by the feed means **290**, and the support boards **400** are moved by the Z-axis feed mechanisms **294**. Thus, as illustrated in FIG. **21**, the tops **402** having a wedge are caused to act on the outer circumference of the ring-shaped reinforcing part **24**, and the wedges of the tops **402** are positioned between the tape **96** and the reinforcing part **24**. In addition, the annular frame **64** is supported by the spheres **404***b* of the frame support parts **404**. Further, the belt conveyor **300** of the discard part **276** is moved from the standby position to the collection position.
(236) Subsequently, irradiation with ultraviolet from the pair of ultraviolet irradiation parts **270** is executed to reduce the adhesive force of the tape **96** sticking to the ring-shaped reinforcing part **24**. In addition, the frame unit U is rotated together with the second raising-lowering table **272** by the motor **284** relative to the separator **274**. Moreover, ionized air is blown from the ionizer **406** toward the frame unit U. With this, the ring-shaped reinforcing part **24** can be separated from the frame unit U as illustrated in FIG. **22**. In addition, static electricity generated when the reinforcing part **24** is separated does not remain on the frame unit U. The reinforcing part **24** that has dropped from the frame unit U is conveyed to the dust box **302** by the belt conveyor **300** and is collected. When the reinforcing part **24** is separated, the separator **274** may be rotated relative to the frame unit U.
(237) (Ring-Free Unit Carrying-Out Step)
(238) After the reinforcing part removal step is executed, a ring-free unit carrying-out step is executed to carry the ring-free unit U′ resulting from the removal of the ring-shaped reinforcing part **24** out from the reinforcing part removing unit **194**.
(239) In the ring-free unit carrying-out step, the belt conveyor **300** of the discard part **276** of the reinforcing part removing unit **194** is first moved from the collection position to the standby

position. Next, the frame holding part **306** of the inversion mechanism **308** (see FIG. **23**) of the ring-free unit carrying-out means **196** is positioned below the ring-free unit U′ held under suction by the second raising-lowering table **272**.

(240) Subsequently, the arm **318** is raised in the state in which the suction pads **326** of the frame holding part **306** are oriented upward, and the suction pads **326** of the frame holding part **306** are brought into contact with the lower surface side of the annular frame **64** of the ring-free unit U′ supported by the second raising-lowering table **272**.

(241) Next, a suction force is generated for the suction pads **326** of the frame holding part **306**, and the annular frame **64** of the ring-free unit U′ is held under suction by the suction pads **326**. Subsequently, the suction holding of the ring-free unit U′ by the second raising-lowering table **272** is released. Thus, the ring-free unit U′ is transferred from the second raising-lowering table **272** of the reinforcing part removing unit **194** to the frame holding part **306** of the ring-free unit carrying-out means **196**.

(242) (Ring-Free Unit Housing Step)

(243) After the ring-free unit carrying-out step is executed, a ring-free unit housing step is executed to house the ring-free unit U′ carried out by the ring-free unit carrying-out means **196**.

(244) In the ring-free unit housing step, the inversion mechanism **308** of the ring-free unit carrying-out means **196** is first inverted upside down, and the ring-free unit U′ held under suction by the frame holding part **306** is inverted upside down. With this, the ring-free unit U′ is located under the frame holding part **306**.

(245) Next, the Y-axis movable member **316** and the arm **318** of the inversion mechanism **308** are moved, and the ring-free unit U′ is brought into contact with the upper surfaces of the pair of support plates **328** of the ring-free unit support part **310**. At this time, the space between the pair of support plates **328** is narrowed by the space adjusting means, and the pair of support plates **328** are in close contact with each other. Subsequently, the suction holding of the ring-free unit U′ by the frame holding part **306** is released to place the ring-free unit U′ on the pair of support plates **328**. Next, slack and wrinkles of the tape **96** caused due to the removal of the reinforcing part **24** are eliminated by actuating the heater mounted on each support plate **328** and heating the tape **96** of the ring-free unit U′. Then, the ring-free unit U′ is held under suction again and is raised by the frame holding part **306**.

(246) Next, after the space between the pair of support plates **328** is enlarged by the space adjusting means, the ring-free unit U′ is placed on the upper surfaces of the support plates **328**. Then, as illustrated in FIG. **25**, the ring-free unit U′ supported by the ring-free unit support part **310** is pushed by the pressing piece **338** of the pushing part **312**, and the ring-free unit U′ is caused to enter the frame cassette **198** placed on the frame cassette table **200** and is housed therein.

(247) The configuration of the present embodiment is as above. In the processing apparatus **2** of the present embodiment, work of integrating the wafer **4** and the annular frame **64** through the tape **96** can automatically be executed even when the wafer **4** has the ring-shaped reinforcing part **24** in a protrusion shape formed in a region on the back surface **4***b* corresponding to the outer circumferential surplus region **20**. In addition, the ring-shaped reinforcing part **24** can automatically be cut and removed from the wafer **4**.

(248) In the present embodiment, description has been made mainly regarding the case in which the ring-shaped reinforcing part **24** is removed. However, in the case of the wafer **4′** without the reinforcing part, there is no need to remove the reinforcing part, and therefore, the processing apparatus **2** can be used without using the reinforcing part removing unit **194** and the ring-free unit carrying-out means **196**.

(249) The present invention is not limited to the details of the above described preferred embodiment. The scope of the invention is defined by the appended claims and all changes and modifications as fall within the equivalence of the scope of the claims are therefore to be embraced by the invention.

Claims

1. A processing apparatus comprising: a wafer cassette table configured to have placed thereon a wafer cassette having multiple wafers housed therein; a wafer carrying-out unit configured to carry a wafer out from the wafer cassette placed on the wafer cassette table; a wafer table configured to support the wafer carried out by the wafer carrying-out unit; a frame housing unit configured to house multiple annular frames therein, each annular frame having an opening part for housing the wafer therein; a frame carrying-out unit configured to carry the annular frame out from the frame housing unit; a frame table configured to support the annular frame carried out by the frame carrying-out unit; a camera configured to image an exposed surface of the wafer, to which the tape of the tape-attached annular frame is to be pressure-bonded by the second tape pressure bonding unit; a first tape pressure bonding unit that is disposed above the frame table and includes a first pressure bonding roller configured to execute pressure bonding of a tape to the annular frame; a tape-attached frame conveying unit configured to convey, to the wafer table, the annular frame to which the tape is pressure-bonded, wherein the tape-attached frame conveying unit is also configured to place the tape-attached annular frame on the wafer table in such a manner that a front surface or a back surface of the wafer supported by the wafer table is positioned in the opening part of the annular frame; a second tape pressure bonding unit that includes a second pressure bonding roller configured to execute pressure bonding of the tape of the tape-attached roller annular frame to the front surface or the back surface of the wafer; frame unit carrying-out means configured to carry, out from the wafer table, a frame unit in which the tape of the tape-attached annular frame and the front surface or the back surface of the wafer are pressure-bonded together by the second tape pressure bonding unit; and a frame cassette table configured to have placed thereon a frame cassette configured to house the frame unit therein; wherein a first heating unit is disposed in one of or both the frame table and the first pressure bonding roller, and a second heating unit is disposed in one of or both the wafer table and the second pressure bonding roller, and either an adhesive tape that has an adhesive layer laid on a sheet or a thermocompression bonding tape that does not have an adhesive layer on a sheet is selectively used as the tape, and when the exposed surface of the wafer supported by the wafer table is the front surface, the camera is configured to acquire an identification displayed on the front surface of the wafer.

2. The processing apparatus according to claim 1, wherein: a ring-shaped reinforcing part having a protrusion shape is formed in a region on the back surface corresponding to an outer circumferential surplus region of the wafer, and the processing apparatus further comprises: a reinforcing part removing unit configured to cut and remove the ring-shaped reinforcing part from the wafer of the frame unit carried out by the frame unit carrying-out means, and ring-free unit carrying-out means configured to carry a ring-free unit resulting from the removal of the ring-shaped reinforcing part out from the reinforcing part removing unit.

3. The processing apparatus according to claim 1, wherein the first tape pressure bonding unit includes: a roll tape support part configured to support a roll tape where the tape that is yet to be used is wound, a tape take-up part that takes up the tape that has been used, a tape pull-out part configured to pull out the tape from the roll tape, the first pressure bonding roller is configured to execute pressure bonding of the tape pulled out to the annular frame, and a cutting part configured to cut the tape protruding to an outer circumference of the annular frame along the annular frame.

4. The processing apparatus according to claim 3, wherein: the tape is a thermocompression bonding tape, and the first heating unit is configured to be actuated to heat one of or both the frame table and the first pressure bonding roller and to execute thermocompression bonding of the thermocompression bonding tape to the annular frame.

5. The processing apparatus according to claim 1, wherein the second tape pressure bonding unit includes: an upper chamber disposed above the wafer table, a lower chamber in which the wafer

table is housed, a raising-lowering mechanism configured to raise and lower the upper chamber and configured to generate a closed state in which the upper chamber is brought into contact with the lower chamber and an opened state in which the upper chamber is separated from the lower chamber, a vacuum part configured to set the upper chamber and the lower chamber to a vacuum state in the closed state, and an opening-to-atmosphere part configured to open the upper chamber and the lower chamber to atmosphere, and, wherein in a state in which the tape of the tape-attached annular frame is positioned to the front surface or the back surface of the wafer supported by the wafer table, the upper chamber and the lower chamber are set to a vacuum state while the closed state is kept through actuating the raising-lowering mechanism, and pressure bonding of the tape of the tape-attached annular frame to the front surface or the back surface of the wafer is configured to be executed by the second pressure bonding roller disposed in the upper chamber.
6. The processing apparatus according to claim 5, wherein, wherein: the tape is a thermocompression bonding tape, and the second heating unit is configured to be actuated to heat one of or both the wafer table and the second pressure bonding roller and to execute thermocompression bonding of the thermocompression bonding tape to the front surface or the back surface of the wafer.
7. The processing apparatus according to claim 5, wherein: the camera is disposed in the upper chamber of the second tape pressure bonding unit, and whether the exposed surface of the wafer supported by the wafer table is the front surface or the back surface is detected before the tape-attached annular frame is conveyed.
8. The processing apparatus according to claim 7, wherein: the camera is configured to detect whether or not the tape is properly pressure-bonded to the wafer, after the pressure bonding of the tape of the tape-attached annular frame to the wafer supported by the wafer table is executed.
9. The processing apparatus according to claim 1, wherein: the frame table is configured to be raised and lowered; the first pressure bonding roller is configured to execute pressure bonding of the tape to the annular frame while the frame is seated upon the frame table; and the frame table is configured to be raised in the vertical direction while the first pressure bonding roller is moved in the horizontal direction such that an elevation angle α formed between the annular frame and the tape extending from the first pressure bonding roller towards a tape take-up part configured to take up the tape that has been used remains constant.
10. A processing apparatus comprising: a wafer cassette table configured to have placed thereon a wafer cassette having multiple wafers housed therein; a wafer carrying-out unit configured to carry a wafer out from the wafer cassette placed on the wafer cassette table; a wafer table configured to support the wafer carried out by the wafer carrying-out unit; a frame housing unit configured to house multiple annular frames therein, each annular frame having an opening part for housing the wafer therein; a frame carrying-out unit configured to carry the annular frame out from the frame housing unit; a frame table configured to support the annular frame carried out by the frame carrying-out unit; a first tape pressure bonding unit that is disposed above the frame table and includes a first pressure bonding roller configured to execute pressure bonding of a tape to the annular frame; a tape-attached frame conveying unit configured to convey, to the wafer table, the annular frame to which the tape is pressure-bonded, wherein the tape-attached frame conveying unit is also configured to place the tape-attached annular frame on the wafer table in such a manner that a front surface or a back surface of the wafer supported by the wafer table is positioned in the opening part of the annular frame; a second tape pressure bonding unit that includes a second pressure bonding roller configured to execute pressure bonding of the tape of the tape-attached annular frame to the front surface or the back surface of the wafer; frame unit carrying-out means configured to carry, out from the wafer table, a frame unit in which the tape of the tape-attached annular frame and the front surface or the back surface of the wafer are pressure-bonded together by the second tape pressure bonding unit; and a frame cassette table configured to have placed thereon a frame cassette configured to house the frame unit therein; wherein a first heating unit is

disposed in one of or both the frame table and the first pressure bonding roller, and a second heating unit is disposed in one of or both the wafer table and the second pressure bonding roller, and either an adhesive tape that has an adhesive layer laid on a sheet or a thermocompression bonding tape that does not have an adhesive layer on a sheet is selectively used as the tape, and wherein the second tape pressure bonding unit includes: an upper chamber disposed above the wafer table, a lower chamber in which the wafer table is housed, a raising-lowering mechanism configured to raise and lower the upper chamber and configured to generate a closed state in which the upper chamber is brought into contact with the lower chamber and an opened state in which the upper chamber is separated from the lower chamber, a vacuum part configured to set the upper chamber and the lower chamber to a vacuum state in the closed state, and an opening-to-atmosphere part configured to open the upper chamber and the lower chamber to atmosphere, and, wherein in a state in which the tape of the tape-attached annular frame is positioned to the front surface or the back surface of the wafer supported by the wafer table, the upper chamber and the lower chamber are set to a vacuum state while the closed state is kept through actuating the raising-lowering mechanism, and pressure bonding of the tape of the tape-attached annular frame to the front surface or the back surface of the wafer is configured to be executed by the second pressure bonding roller disposed in the upper chamber, and wherein a camera is disposed in the upper chamber of the second tape pressure bonding unit, and whether an exposed surface of the wafer supported by the wafer table is the front surface or the back surface is detected before the tape-attached annular frame is conveyed.
11. The processing apparatus according to claim 10, wherein: when the exposed surface of the wafer supported by the wafer table is the front surface, the camera is configured to acquire an identification displayed on the front surface of the wafer.
12. The processing apparatus according to claim 10, wherein: the camera is configured to detect whether or not the tape is properly pressure-bonded to the wafer, after the pressure bonding of the tape of the tape-attached annular frame to the wafer supported by the wafer table is executed.
13. A processing apparatus comprising: a wafer cassette table configured to have placed thereon a wafer cassette having multiple wafers housed therein; a wafer carrying-out unit configured to carry a wafer out from the wafer cassette placed on the wafer cassette table; a wafer table configured to support the wafer carried out by the wafer carrying-out unit; a frame housing unit configured to house multiple annular frames therein, each annular frame having an opening part for housing the wafer therein; a frame carrying-out unit configured to carry the annular frame out from the frame housing unit; a frame table configured to support the annular frame carried out by the frame carrying-out unit; a first tape pressure bonding unit that is disposed above the frame table and includes a first pressure bonding roller configured to execute pressure bonding of a tape to the annular frame; a tape-attached frame conveying unit configured to convey, to the wafer table, the annular frame to which the tape is pressure-bonded, wherein the tape-attached frame conveying unit is also configured to place the tape-attached annular frame on the wafer table in such a manner that a front surface or a back surface of the wafer supported by the wafer table is positioned in the opening part of the annular frame; a second tape pressure bonding unit that includes a second pressure bonding roller configured to execute pressure bonding of the tape of the tape-attached annular frame to the front surface or the back surface of the wafer; frame unit carrying-out means configured to carry, out from the wafer table, a frame unit in which the tape of the tape-attached annular frame and the front surface or the back surface of the wafer are pressure-bonded together by the second tape pressure bonding unit; and a frame cassette table configured to have placed thereon a frame cassette configured to house the frame unit therein; wherein a first heating unit is disposed in one of or both the frame table and the first pressure bonding roller, and a second heating unit is disposed in one of or both the wafer table and the second pressure bonding roller, and either an adhesive tape that has an adhesive layer laid on a sheet or a thermocompression bonding tape that does not have an adhesive layer on a sheet is selectively used as the tape, wherein

a ring-shaped reinforcing part having a protrusion shape is formed in a region on the back surface corresponding to an outer circumferential surplus region of the wafer, and the processing apparatus further comprises: a reinforcing part removing unit configured to cut and remove the ring-shaped reinforcing part from the wafer of the frame unit carried out by the frame unit carrying-out means, and ring-free unit carrying-out means configured to carry a ring-free unit resulting from the removal of the ring-shaped reinforcing part out from the reinforcing part removing unit, and wherein the reinforcing part removing unit further includes a separating part that is configured to separate the ring-shaped reinforcing part from the remainder of the wafer via a plurality tops that each have a wedge shape and that are configured and arranged to be brought into contact with an outer peripheral surface of the reinforcing part.

14. The processing apparatus according to claim 13, wherein: a ring-shaped reinforcing part having a protrusion shape is formed in a region on the back surface corresponding to an outer circumferential surplus region of the wafer, and the processing apparatus further comprises: a reinforcing part removing unit configured to cut and remove the ring-shaped reinforcing part from the wafer of the frame unit carried out by the frame unit carrying-out means, and ring-free unit carrying-out means configured to carry a ring-free unit resulting from the removal of the ring-shaped reinforcing part out from the reinforcing part removing unit.